



Transcend Network Automation Suite

Transcend Network Management System (TNMS)

Release 9.1

Installation Manual (IMN, Linux)

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1. Preface

Transcend Network Management System (TNMS) is a Nokia standalone application that manages the physical structure of the transport network as well as the logical end-to-end connections and services.

The TNMS Installation Manual describes the installation and uninstallation procedures of TNMS, with a TNMS Server running on Linux.

The contents of this manual apply only to TNMS 9.1.

1.1. Intended audience

This document is intended for TNMS installers and administrators with knowledge of UNIX and Linux.

1.2. Symbols and Conventions

The following symbols and mark-up conventions are used in this document:

Table 1: List of symbols and conventions

Representation	Meaning
 DANGER!	A safety message indicates a dangerous situation where personal injury is possible. The keywords denote hazard levels with the following meaning:
 WARNING!	 DANGER! Indicates a hazardous situation which, if not avoided, will result in death or serious (irreversible) personal injury.
 CAUTION!	 WARNING! Indicates a hazardous situation which, if not avoided, could result in death or serious (irreversible) personal injury.
	 CAUTION! Indicates a hazardous situation which, if not avoided, may result in minor or moderate (reversible) personal injury.
 NOTICE:	A property damage message indicates a hazard that may result in equipment damage, data loss, traffic interruption, and so on.
 Note:	A note provides important information related to the topic, for example, not obvious exceptions to a rule or side effects.
 Tip:	A tip provides additional information related to the topic which is not essential in the context, but given for convenience.
Bold	• All names of graphical user interface (GUI) objects, such as windows, field names, buttons, and so on. Example: Select the Full Screen check box and press OK. • Terms and abbreviations which are linked to an entry in the glossary and list of abbreviations respectively. • Important key words.

Table 1: List of symbols and conventions (continued)

Representation	Meaning
<i>Italic</i>	- Files, folders, and file system paths. Example: <code>/usr/etc/sbin/ftpd.exe</code> - Emphasized words. - Document titles. Example: Refer to <i>WebGUI User Guide</i>
typewriter	- Input to be typed in a command line or a GUI field. Examples: <code>ping -t 192.168.0.1</code> Enter <code>World</code> in the Domain field. - Output from a command, error messages, content of a status line, and so on. - File content, such as program sources, scripts, logs, and settings.
<angle brackets>	Placeholders, for example as part of a file name or field value. Examples: <code><picture name>.png</code> or <code><ip address>:<port number></code>
[square brackets]	A key to be pressed on a PC keyboard, for example [F11]. Keys to be pressed simultaneously are concatenated with a "+" sign, for example [CTRL]+[ALT]+[DEL]. Keys to be pressed one after another are concatenated with spaces, for example [ESC] [SPACE] [M].
>	The greater than symbol ">" is used to concatenate a series of GUI items in order to depict a <i>GUI path</i>. This is an abridged presentation of a procedure to be carried out in order to perform an action or display a window or dialog box. Examples: A simple menu path: File > Save as ... A more complex GUI path: > Main window > File menu > Change Password command > Change Password dialog box
x (in card names)	For convenience, card names are sometimes listed with a lower case x variable, in order to concisely represent multiple cards. Example: CHMx (is to be interpreted as CHM1 and CHM2)
	Please read the relevant documents carefully before use.

Screenshots of the graphical user interface are examples only to illustrate principles. This especially applies to a software version number visible in a screenshot.

1.3. TNMS and Components Documentation

The operational documents listed below are available in the **TNMS_Prerequisites > Documentation** folder, `<Product_Installation_Folder>\TNMS\client\help` or via the TNMS help menu (select manuals).

Transcend NMS Documentation

- *TNMS Administration Manual*
- *TNMS Communication Matrix*
- *TNMS Dimensioning Guidelines*
- *TNMS Installation Manual Linux*
- *TNMS Installation Manual Windows*
- *TNMS List of Configurable Parameters and Default Values*
- *TNMS System Description and Operation Guidelines*
- *TNMS Troubleshooting Manual*
- *TNMS Upgrade Manual*
- *TNMS Online Help* (via F1 or the help button in the TNMS Client GUI)
- *Transcend Client Help*

Transcend Node Manager Documentation

- *TNMS Node Manager Documentation Update*
- *TNMS 5500 Node Manager User Manual*
- *TNMS 7090 Node Manager User Manual*
- *TNMS 7100 and mTera Node Manager User Manual*

1.4. Other Documents

Legacy products and Network Elements

This manual concerns TNMS only. For more detailed information on other legacy products or the managed network elements (NEs), see the corresponding documentation.

Release notes

Where applicable, contains installation hints, patch descriptions, list of supported NEs, list of supported cards and any relevant last minute information.

1.5. Documentation Feedback

To comment on this document, go to the [Online Comment Form](https://documentation.nokia.com/comments/) (<https://documentation.nokia.com/comments/>) or e-mail your comments to the [Comments Hotline](mailto:comments@nokia.com) (<mailto:comments@nokia.com>).

1.6. History of Changes

This chapter describes the main changes for the current document and since the last version.

Table 2: History of Changes

Document number	Issue date	Remarks
A50023-K2268-X040-76D1	August 2026	Chapter Setting Up the FTP Service (p. 61) was updated.
		Chapter TNMS Server and Mediation Installation (p. 43) was updated.
		Chapter TNMS Frontend Server Installation (p. 55) was updated.

Table 2: History of Changes (continued)

Document number	Issue date	Remarks
		Chapter Hardware Requirements (p. 13) was updated.
		Chapter Disk Configuration (p. 19) was updated.
		Chapter TNMS Installer Integrity Validation (p. 42) was updated.
		Chapter Glossary (p. 76) was updated.
		Chapter Linux 8.X (p. 23) was updated.
		Chapter Linux 9.X (p. 25) was updated.
		Chapter TNMS Database Installation and Configuration (p. 37) was updated.
		Chapter Virtualization (p. 15) was updated.
		Chapter Before you Begin (p. 11) was updated.

2. Preparation

Before installing TNMS, read the following sections to ensure that all requirements are met.



Tip: To avoid unnecessary usage of disk space, it is recommended to:

- Store the installation *zip* files in a partition other than the one used for the software installation.
- Delete the unzipped *zip* files after software installation.



Note: Configuration is done with standard Unix tools provided with Linux. Check their available man files for detailed information.



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. This chapter contains instructions for RHEL installation only. All subsequent Linux instructions, hardening and commands in this manual are also based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

2.1. Before you Begin

Complete the following steps before installing:

- If Hot-Standby is planned to be used, consider using two redundant communication links between Primary and Secondary sites. Enable redundancy on the ethernet interfaces by configuring NIC bonding.
- Check the system requirements.
- Determine the file system to be used, the partition to be used by the installation and the components to install.
- Determine how the network, IP addresses and TCP/IP name management will be handled.
- Ensure that the host IP addresses are static. Do not use DHCP dynamic addresses.
- Verify that the machine where TNMS will be installed does not already have any version of TNMS, Transcend eDNA Frontend Server or Transcend Controller installed (only one role per machine is permitted).
For upgrades of installed versions, see the *TNMS Upgrade Manual*.
- Create a management-console (ILO) account to HP Server.
- It is mandatory to install the database in the TNMS Server machine.
- If TMF CORBA NBI is required contact Technical Support before proceeding.
- Consult the Dimensioning Heuristics chapter, in the *TNMS Dimensioning Guidelines* manual, for dimensioning configurations of systems:
 - that exceed the certified system parameters and/or
 - where default system parameters are inadequate to process the required workloads.

2.1.1. TNMS Installation Checklist

In order to take note and keep track of relevant parameters and configurations, it is recommended that the attached *TNMS Installation Checklist (Linux)* spreadsheet is filled in when installing your Operating System, Pre-requisites Software and TNMS.

! **NOTICE:** This file is confidential and the information that is filled in is highly sensitive in nature.

The *TNMS Installation Checklist* is a basic template structured with the following sections:

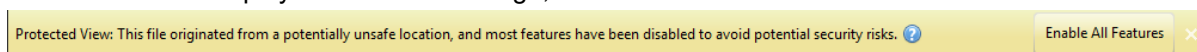
- Read Me First
- Table of Contents
- System Configuration
- Initial Configuration
- Software Pre-requisites
- TNMS Installation
- Final Configuration
- General Notes
- Print Instructions

The configuration settings and parameter fields that are provided can be edited and expanded upon, based on the operator's needs. The information that is filled out in the *TNMS Installation Checklist (Linux)* should be saved so that it can be referred back to whenever needed.

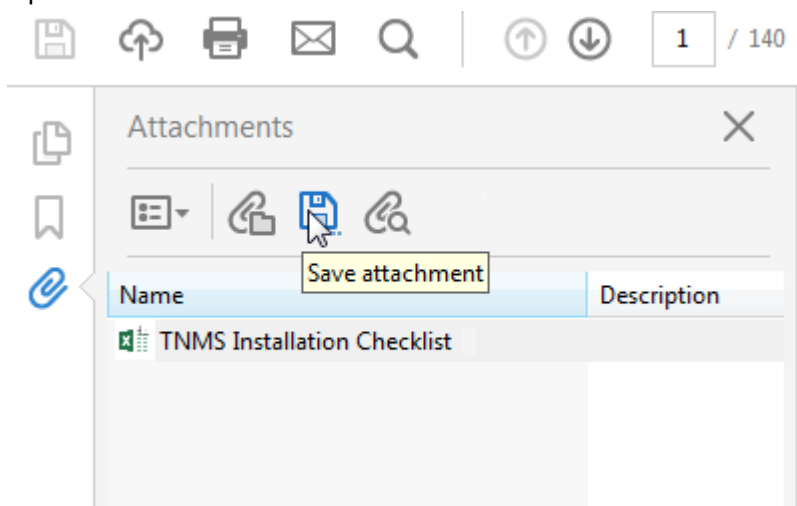
Saving the TNMS Installation Checklist

To save the *TNMS Installation Checklist (Linux)*:

- If Adobe Reader displays the below message, click **Enable All Features**



- Open the **Attachments** tab situated below the **Bookmarks** tab.



- Select the TNMS Installation Checklist file and click the **Save Attachment** icon.

- Choose a directory to save the *TNMS Installation Checklist (Linux)* file.



NOTICE: When saving this file, the following security measures are strongly advised:

- Save the file on an encrypted storage or disk (internal or external).
- Restrict storage or disk access to the internal network (no Internet access).
- Restrict storage or disk access to Administrators and authorized authenticated users.
- Password protect the file from unauthorized users.

The checklist spreadsheet is now ready to be populated with your configuration settings, parameters and notes.

2.2. Installation and Cabling

Before installation, make sure that the components are not damaged in any way, and the delivery is complete and in accordance with the delivery units specified in the delivery note (hardware, software, licenses and documentation).

Do not start the installation without checking if:

- The Client and Server are cabled and set up according to project specifications.
- Uninterruptible power supply is available.
- Any other applicable recommendations are observed.

2.3. Hardware Requirements

The table below gives an overview of the hardware requirements for deploying TNMS.

The Hardware Requirements listed in [Table 3: Hardware requirements for installations of TNMS \(p. 14\)](#) should suffice for most deployments, however TNMS may require different specifications depending on parameters such as network architecture (number of TNMS Clients) or operation policies (backup, logs).

The final hardware specifications and configuration must be planned specifically for each customer. Contact Technical Sales for more information.



Tip: It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)* spreadsheet.

For more information regarding this checklist refer to chapter [TNMS Installation Checklist \(p. 12\)](#)

In all recommended configurations, TNMS Server and TNMS Mediation(s) must be installed on the same machine.

Refer to chapter [TNMS Special Deployment \(p. 50\)](#) for instructions about TNMS Server and TNMS Mediation installations.

Table 3: Hardware requirements for installations of TNMS

Components		Hardware configurations			
Software	Hardware	Small	Small Plus	Medium	Large
TNMS Server + TNMS Mediation	CPU	8C/16T/16 vcpus Max Turbo 3.7 GHz	16C/32T/32 vcpus Max Turbo 3.7 GHz	32C/64T/64 vcpus Max Turbo 3.7 GHz	80C/160T/160 vcpus Max Turbo 3.7 GHz
	RAM	16 GB DDR4-2666 ¹⁾	32 GB DDR4-2666 ¹⁾	64 GB DDR4-2666 ¹⁾	256 GB DDR4-2666 ¹⁾
	Disk	(1x) 1 TB HDD	(2x) 600 GB SSD	(2x) 1 TB SSD	(1x) 960 GB SSD + (3x) 1.2 TB SSD
	Disks with redundancy	(2x) 1 TB HD (Raid-1)	(4x) 600 GB SSD (Raid-1)	(4x) 1 TB SSD (Raid-1)	(2x) 960 GB SSD (Raid-1) + (6x) 1.2 TB SSD (Raid-6)
TNMS FE Server	CPU	24C/48T/48 vcpus Max Turbo 3.7 GHz			
	RAM	100 GB DDR4-2666 ¹⁾			
	Disk	(1x) 400 GB SSD			
	Disks with redundancy	(2x) 400 GB SSD (Raid-1)			
TNMS Client (Windows)	CPU	4 core / 8 threads / 8 vcpus 3.20GHz Recommended GPUs: • Intel Core i3/i5/i7-3000 series or newer. • AMD A4/A6/A8/A10 5000 series or newer. • GPUs with support for OpenGL ES3.0 or newer.			
	RAM	8 GB DDR4-2666 ¹⁾			
	Disk	500 GB			

¹⁾RAM: DDR4-2666 is the minimum requirement. Higher speeds or DDR5 can also be used.



Note: In all configurations, AMD64/x86_64 processor architectures must be used.



Note: Usage of the former disk configuration with 600GB disks in a Medium configuration, is only supported if the number of 15min PMPs is under 20000.



NOTICE: Transcend eDNA and Node Manager are supported in a medium or large configuration only.

Contact Technical Support for more information.

 **Note:** Small configurations in production environments require pre-approval by a TNMS Product Line Manager, and are only supported for networks with:

- Only one of the following mediators:
 - Generic Mediator
 - MultiVendor Mediator
- The bare minimum of TNMS components and NBIs installed, only if strictly necessary.

 **NOTICE:** Node Manager and eDNA are not supported in a Small configuration in a production environment.

 **NOTICE:** If Transcend Controller is to be included in the network, utilizing the TNMS database, the TNMS Server machine requires 256 GB of extra for the Medium configuration. The extra disk space is required in the ora2 filesystem.

 **Note:** To enhance TNMS machine High Availability, use hardware with more Ethernet physical ports to support link aggregation. The aggregation of Ethernet ports can provide higher system throughput and availability by using load balancing and fail over mechanisms, respectively.

To perform link aggregation, consider using the Network Bonding solution of the RHEL OS.

2.3.1. Virtualization

The current version of TNMS has been tested on several virtualization technologies. There should be no technical reason to impede the correct functioning of the TNMS server on a virtual machine, if all system requirements (hardware requirements, configurations and supported operating systems) described in this document are followed.

 **Note:** The system requirements outlined in this document apply to the specifications of the virtual machine not the virtual host/hypervisor.

2.4. Supported Operating Systems

Throughout this and the following chapters the designation of the several operating systems is often abbreviated to allow for better readability. Always refer to the current section for the exact versions supported for TNMS.

 **Tip:** It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)*. For more information regarding this checklist refer to chapter [TNMS Installation Checklist \(p. 12\)](#) spreadsheet.

 **Note:** Refer to the *Release Notes Appendix, Licensed Software* chapter for an updated list of supported Linux versions for the Server machine.

Note: For networks that include 7100 Nano FP13.0.1 and earlier NEs, contact Technical Support.

Note: Minimum display resolution is 1024x768.

Note: English must be the Operating System language for the Server machine. If you must use another language, contact Nokia.

2.5. Software Prerequisites by Component

The following table describes which software is required for each component. Attend to the fact that the table also shows the order in which the components should be installed. After installing the operating system, the system should be commissioned as follows:

Table 4: TNMS software prerequisites and their installation sequence

Software	Server	Client
Dedicated PDF Reader	-	Mandatory to open the manuals.
Software Prerequisites Installation (p. 37) Oracle19c installation package	Mandatory	-

2.6. Latency and Bandwidth Requirements

Verify that your network complies with the values in the following tables:

Table 5: Minimum bandwidth

From	To			
	NEs	TNMS Mediation	TNMS Server	TNMS Clients
NEs	n/a	10 Mb/s	n/a	n/a
TNMS Mediation	100 Mb/s	n/a	100 Mb/s	n/a
TNMS Server	n/a	100 Mb/s	50 Mb/s (to Standby Server)	100 Mb/s
TNMS Clients	n/a	n/a	10 Mb/s	n/a

Table 6: Maximum latency (Round-trip delay time)

From	To			
	NEs	TNMS Mediation	TNMS Server	TNMS Clients
NEs	n/a	50 ms	n/a	n/a
TNMS Mediation	50 ms	n/a	50 ms	n/a
TNMS Server	n/a	50 ms	50 ms (to Standby Server)	50 ms
TNMS Clients	n/a	n/a	50 ms	n/a

2.7. Installation Software

Nokia delivers the following software:

- TNMS Server for Linux.
- TNMS Server and Mediation for Linux.
- TNMS Mediation for Linux.
- TNMS Client for Windows.

 **Note:** The versions of the software to install must be the ones from the original product.

 **Note:** Only required software must be installed on either the Server or Client machines. Installing additional software may cause TNMS not to function correctly.

 **Note:** Read the *Release Notes* before installing TNMS.

2.8. Configuring the BIOS

This chapter describes the recommended configurations for the system's BIOS.

 **NOTICE:** The instructions in this chapter refer to HP machines with iLO 6 and may differ depending on hardware configurations.

To access the BIOS, boot the machine and press **F9** at the startup screen.

- Disable the boot from the network:
Go to **System Configuration > BIOS (RBSU) > Network Options > Network Boot Options > PCIe Slot Network Boot** and set all Network Interface Cards to **Disabled**.
- Processor options:
 - Go to **System Configuration > BIOS (RBSU) > Virtualization Options > Intel Virtualization Technology** (intel VT) and set to **Disabled**.
 - Go to **System Configuration > BIOS (RBSU) > Virtualization Options > Intel VT-d**, and set to **Disabled**.
- Power management options:
 - Go to **System Configuration > BIOS/Platform Configuration (RBSU)** and in the **Workload Profiles dropdown** Select **Transactional Application Processing**.
Press F12 to apply the changes and exit the BIOS.

Once the reboot is complete, go to **Power & Thermal > Power Settings** tab. If the **Power Regulator Setting** is not set to **Static High Performance Mode**, then select it and then click **Apply**. Reboot if requested.

3. TNMS Server Operating System Configuration

This chapter describes the procedures to install and configure the operating system of the TNMS Server. Before installing the server operating system, you must create and configure the logical drive where Linux OS will be installed.

The following chapter applies to the recommended small, small plus, medium and large hardware configurations only and these steps may differ in case you have any other hardware configurations.

 NOTICE: Only fresh installations are covered

The Server software must either have been uninstalled from or never have been installed on, the machine in question.

Perform the server installation **in the order and strictly as prescribed**.

Failure to do so will cause the Server to malfunction.

 **Note:** The screen output in the current manual, while generally correct, is given as an example. Depending on your particular configuration, actual output may differ.

 **NOTICE:** The instructions in this chapter refer to HP machines and may differ for other hardware configurations.

 **Tip:** It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)* spreadsheet.
For more information regarding this checklist refer to chapter [TNMS Installation Checklist \(p. 12\)](#).

 **Note:** TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

3.1. Integrated Lights-Out (iLO 6) Management Console

This chapter describes how to operate the Integrated Lights-Out (iLO) management console. This console is used to access the server machine and for administration purposes. Refer to the iLO specific documentation for further information.

Accessing the Integrated Remote Console

Use the following information to access the console:

1. Address: `https://<machine IP>`
2. Username: `<user>`
3. Password: `<password>`
4. In the left panel tree, expand Information > Overview, and in Integrated Remote Console, click the **.NET** link.

3.2. Disk Configuration

It is recommended that you configure a RAID 1 for the disks where the operating systems will be installed.

While booting the machine, proceed as follows:

- Step 1.** When the **Press any key to view Option ROM messages** appears, click **ENTER**.
- Step 2.** When the internal controller displays the message **Press F10 to enter the Intelligent Provisioning**.
- Step 3.** In the Intelligent Provisioning **Performance Maintenance** page, select **Intelligent Storage Configuration**.
- Step 4.** In the **Intelligent Storage Configuration > Logical Drives** tab, select **Create Array**.
- Step 5.** Using the default settings, create the RAID 1 configuration with the two available hard drives.

 **Note:** Select the appropriate RAID configuration and the layout of disks and file system as both will be requested later on.

 **Note:** It is also possible to configure additional mount points (for example, to comply with CIS Hardening specifications) from the disk space reserved to the *root* File System. For more information refer to the *CIS Hardening Checklist* available in the 9.1 **TNMS_Prerequisites > Documentation** folder.

Table 7: Logical Disk Configuration

Logical Name	TNMS FE Server Configuration		Small Configuration		Small Plus Configuration		Medium Configuration		Large Configuration	
	RAID Level	Number of Disks	RAID Level	Number of Disks	RAID Level	Number of Disks	RAID Level	Number of Disks	RAID Level	Number of Disks
sda	1	2	1	2	1	2	1	2	1	2
sdb	-	-	-	-	1	2	1	2	6	6

Table 8: File system and disk space size

Volume	Mount Point	File System	TNMS FE Server Configuration (GB)	Small Configuration (GB)	Small Plus Configuration (GB)	Medium Configuration (GB)	Large Configuration (GB)
/dev/sda	/	ext4 or xfs	150	940	150	150	170
/dev/sda		swap	100	60	100	100	130
/dev/sda	/tmp	ext4 or xfs	50	-	50	50	60
/dev/sda	/oradata/ora1	ext4 or xfs	-	-	300	700	600

Table 8: File system and disk space size (continued)

Volume	Mount Point	File System	TNMS FE Server Configuration (GB)	Small Configuration (GB)	Small Plus Configuration (GB)	Medium Configuration (GB)	Large Configuration (GB)
/dev/sdb	/oradata/ora2	ext4 or xfs	-	-	300	500	2500
/dev/sdb	/oradata/ora3	ext4 or xfs	-	-	150	250	1500
/dev/sda	/nokia	ext4 or xfs	100	-	-	-	-
/dev/sdb	/nokia	ext4 or xfs	-	-	150	250	800



Note: File system formats (ext4 or xfs) must be the same in all partitions, except for the swap.

3.3. Linux Installation



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. This chapter contains instructions for RHEL installation only. All subsequent Linux instructions, hardening and commands in this manual are also based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

3.3.1. Internet Protocol Address Standards

TNMS supports different standards in Internet Protocol (IP) addresses and can migrate from an IPv4 Data Communication Network to an IPv6 infrastructure.

The user configures the NE to work with IPv4, IPv6 or both, depending on the capabilities of the NE. For example, a user can specify a unique IPv6 address as an alternative to IPv4 addresses in a dual-stack infrastructure, if the NE supports IPv6.

IPv4

IPv4 addresses are 32-bit, numeric dotted decimal notation addresses, for example: 192.159.252.76.

IPv6

IPv6 addresses are 128-bit, Hexadecimal notation (0-9 and A-F) addresses, for example: 3FFE:F200:0234:AB00:0123:4567:8901:ABCD.

All fields in TNMS that support IPv6 addresses:

- Automatically validate the formatting of IPv6 address.
- Automatically validate the special address restrictions.

Table 9: IPv6 Special Address Restrictions:

Address Block (CIDR)	First Address	Last Address	Number of Addresses	Purposes
::/128	::	::	1	Unspecified address.

Table 9: IPv6 Special Address Restrictions: (continued)

Address Block (CIDR)	First Address	Last Address	Number of Addresses	Purposes
fe80::/10	fe80::	febf:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2118	Link-local address
ff00::/8	ff00::	ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff	2120	Multicast address.

3.3.2. Installing Linux



Note: The host IP addresses are static. Do not use DHCP dynamic addresses.

3.3.2.1. Installing Linux 8.X or 9.X



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

Linux Red Hat Enterprise Linux is to be installed on the required hardware (see [Table 3: Hardware requirements for installations of TNMS \(p. 14\)](#)).

To install Linux follow the steps below. Use the arrow keys to move up and down, and *Enter* to select the desired option:

- Step 1.** Start the machine with the Linux Installation DVD in the DVD-ROM drive.
The Linux installation welcome screen is displayed
- Step 2.** Select **Test this media & install Red Hat Enterprise Linux** .
The disk currently in the drive will be tested for possible errors.
If the test is successful, the installation begins.
- Step 3.** In the language selection step, select for example **English (United States)**.
Click **Continue**.
The Installation Summary is displayed.
- Step 4.** In the **Installation Summary** click on **Time & Date**.
Step 4a. In the **Time & Date** step select your region, city, date and time.
Click **Done** to return to the Installation Summary.
- Step 5.** In the **Installation Summary** click on **Software Selection**.
Step 5a. In the **Software Selection** step select **Server with GUI** as Base Environment.
Click **Done** to return to the Installation Summary.
- Step 6.** In the **Installation Summary** click on **Installation Destination**.
Step 6a. In the **Installation Destination** step select the appropriate disks and select **Custom** under **Storage Configuration** Click **Done**.
Step 6b. In the **Manual Partitioning** step configure the partitions according to:
 - [Table 7: Logical Disk Configuration \(p. 19\)](#)
 - [Table 8: File system and disk space size \(p. 19\)](#)
 Click **Done**.

Step 6c. In the **Summary of Changes** review the partitioning setup you made and click **Accept Changes**.



Note: No changes to the disks will be made until you begin the installation (in step 8 below)

Click **Done** to return to the Installation Summary.

Step 7. In the **Installation Summary** click on **Network and Host Name**.

Step 7a. In the **Network and Host Name** step select the network interface card and turn the switch in the upper right corner of the window on.

Enter the host name in the field **Host name**.

Click **Configure** to edit the settings of the selected network interface card.

Step 7b. In the **Editing <network interface card name>** window click on the **IPv4 Settings** tab.

Step 7c. Select the **Method: Manual** and enter your network settings.

Click **Save** and then **Done**.

Step 8. In the **Configuration** screen, click on **Root password**.

a. In the **Root password** enter a password and reenter it.

b. Click **Done** to return to the Configuration screen.

Step 9. In the **Installation Summary**, click on **Begin Installation**.

The **Configuration** screen is displayed with a progress bar showing the installation progress.

Step 10. After the installation finishes, click **Reboot**.

Step 11. In the **Initial Setup** step, press 1 to select License Information.

Step 12. In the **License Information** step, select **I accept the License Agreement**.

Click **Done**.

Step 13. Press **Finish Configuration**.

Step 14. The **Welcome** step is displayed.

Click **Next**.

Step 15. In the **Privacy** step, turn off the **Location Services**.

Click **Next**.

Step 16. In the **Connect Your Online Accounts** step click **Skip**.

Step 17. In the **About You** step add the user details. For example:

- Full Name: System Administrator
- Username: SysAdmin



Note: "tnms" is not a valid Username.

Click **Next**.

Step 18. In the **Set a password** step enter and reenter the user password.

Click **Next**.

Step 19. In the final step click **Start using Red Hat Enterprise Linux Server**.

The Red Hat Enterprise Linux Server installation is complete.

3.3.3. Configuring Database Filesystems

 **Note:** This is only applicable to ext4 filesystems.

For Small Plus, Medium and Large configurations run the following commands to configure the database system:

Step 1. Login as *root*.

Step 2. Execute:

```
# df | grep oradata | cut -d ' ' -f1
```

The output of this command will show the device names for the database data directories. For example:

```
/dev/sda3
```

```
/dev/sda1
```

```
/dev/sda2
```

Step 3. Execute the following command for each device listed in step 2:

```
# tune2fs -m0 device_name
```

For example:

```
# tune2fs -m0 /dev/sda3
```

```
# tune2fs -m0 /dev/sdb1
```

```
# tune2fs -m0 /dev/sdb2
```

3.3.4. Creating Additional Directories

The following steps are only required for the TNMS Small Configuration. The folders should already exist for the Small Plus, Medium and Large Configurations.

To create the additional directories, do as follows:

Step 1. Login as *root*.

Step 2. As *root* create the following directories to support TNMS.

Issue:

```
# mkdir -p /oradata/ora1
```

```
# mkdir -p /oradata/ora2
```

```
# mkdir -p /oradata/ora3
```

3.3.5. Installing Additional Packages

Some additional operating system packages need to be installed. To do so follow the steps below.

 **Note:** For offline installations of Linux, these packages must be downloaded and available in the TNMS Server Machine before proceeding.

3.3.5.1. Linux 8.X

Add repositories and install additional operating system packages:

1. Add the EPEL repository.

For example as root issue:

```
# dnf -y install  
https://dl.fedoraproject.org/pub/epel/epel-release-latest-8.noarch.rpm
```

2. If ZTC is installed, add the KEA Repository.

For example, as root issue:

```
# wget https://dl.cloudsmith.io/public/isc/kea-2-6/setup.rpm.sh  
# chmod +x setup.rpm.sh  
# ./setup.rpm.sh
```

3. Download and install the packages below. This can be done manually (with rpm package manager) or automatically (by configuring a package manager).

```
alsa-lib*  
atk*  
at-spi2-atk*  
at-spi2-core*  
attr  
bash*  
bc  
ca-certificates*  
cairo*  
chkconfig*  
cups-libs*  
dbus-libs*  
elfutils-libelf-devel  
expat*  
fontconfig-devel  
ftp (Install only if required by legacy NEs; refer to NE Release Notes)  
gcc  
gcc-c++  
glib2*  
glibc*  
gtk3*  
isc-kea-dhcp4-2.6.3 (only if ZTC is installed)  
jemalloc  
ksh  
libaio-devel  
libnsl*2*  
libnsl.i686 (For networks with 5500 NEs)  
libstdc++.i686 (For networks with 5500 NEs)
```

```
libXext*
libXi*
libX11*
libXrender*
libXcomposite*
libXdamage*
libXfixes*
libXtst
libXrandr*
libcurl*
libdrm*
liberation-fonts*
libgcc*
libxcb*
libxkbcommon*
mesa-libgbm*
libzip
make
nspr*
nss*
nss-util*
pango*
psmisc
protobuf (only if ZTC is installed)
sysstat
vsftpd (Install only if required by legacy NEs; refer to NE Release Notes)
vulkan-loader*
wget*
xdg-utils*
xorg-x11-server-Xvfb*
zlib.i686 (For networks with 5500 NEs)
```

* For networks that include eDNA.

3.3.5.2. Linux 9.X

Add repositories and install additional operating system packages:

1. Add the EPEL repository.

For example, as *root* issue: `# dnf -y install`

`https://dl.fedoraproject.org/pub/epel/epel-release-latest-9.noarch.rpm`

2. If ZTC is installed, add the KEA Repository.

For example, as root issue:

```
# wget https://dl.cloudsmith.io/public/isc/kea-2-6/setup.rpm.sh
# chmod +x setup.rpm.sh
# ./setup.rpm.sh
```

3. Download and install the packages below. This can be done manually (with rpm package manager) or automatically (by configuring a package manager).

```
alsa-lib*
alternatives*
at-spi2-atk*
at-spi2-core*
atk*
attr
bash*
bc
binutils
ca-certificates*
cairo*
cups-libs*
compat-openssl11
chkconfig
dbus-libs*
elfutils-libelf
elfutils-libelf-devel
expat*
fontconfig
fontconfig-devel
ftp (Install only if required by legacy NEs; refer to NE Release Notes)
glibc
glibc-devel
glib2*
gtk3*
isc-kea-dhcp4-2.6.3 (only if ZTC is installed)
jemalloc
ksh
libaio
libasan
libcurl*
libdrm*
liberation-fonts*
```

liblsan
libX11
libXau
libXi
libXrender
libXtst
libxcrypt-compat
libgcc
libibverbs
libnsl*2*
librdmacm
libstdc++
libxcb
libxkbcommon*
libxcomposite*
libXdamage*
libvirt-libs
libnsl.i686 (For networks with 5500 NEs)
libstdc++.i686 (For networks with 5500 NEs)
libXext*
libXfixes*
libXrandr*
make
mesa-libgbm*
net-tools
nspr*
nss*
nss-util*
pango*
policycoreutils
policycoreutils-python-utils
polkit
protobuf (only if ZTC is installed)
smartmontools
sysstat
vsftpd (Install only if required by legacy NEs; refer to NE *Release Notes*)
vulkan-loader*

```
wget*  
xdg-utils*  
xorg-x11-server-Xvfb*  
zlib.i686 (For networks with 5500 NEs)
```

* For networks that include eDNA.

3.3.6. Checking the System Locale

After installing the OS, ensure the system locale is correct by executing the following command:

1. Login as *root*.
2. Issue:

```
#locale
```

The following is an example of the command output:

```
LANG=pt_PT.UTF-8  
LC_CTYPE="pt_PT.UTF-8"  
LC_NUMERIC="pt_PT.UTF-8"  
LC_TIME="pt_PT.UTF-8"  
LC_COLLATE="pt_PT.UTF-8"  
LC_MONETARY="pt_PT.UTF-8"  
LC_MESSAGES="pt_PT.UTF-8"  
LC_PAPER="pt_PT.UTF-8"  
LC_NAME="pt_PT.UTF-8"  
LC_ADDRESS="pt_PT.UTF-8"  
LC_TELEPHONE="pt_PT.UTF-8"  
LC_MEASUREMENT="pt_PT.UTF-8"  
LC_IDENTIFICATION="pt_PT.UTF-8"  
LC_ALL=
```

Make sure the `LANG` variable is set to a UTF-8 encoding and that the `LC_ALL` variable is not set.

In case the `LANG` variable is not set, or the `LC_ALL` is set, edit the Linux system configuration file for the locale information:

```
/etc/sysconfig/i18n.
```

For example:

```
LANG="en_US.UTF-8"
```

If no UTF-8 locale is defined in `LANG`, `LC_CTYPE` or `LC_ALL` in the output above, check there are UTF-8 encodings for your language:

```
# locale -a | grep -i utf
```

If no output is returned, install the language packs for your region.

4. Initial System Configuration



Tip: It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)* spreadsheet.

For more information regarding this checklist refer to chapter [TNMS Installation Checklist \(p. 12\)](#).



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

4.1. System Hosts

TNMS requires the proper configuration of the *hosts* file (*/etc/hosts*) in every Server machine. Each host file must also include a *localhost* entry.

If the network also includes Transcend eDNA, all Transcend eDNA (TNMS Server and TNMS FE Server) machines must be listed in each TNMS machine hosts file.

The configuration of *hosts* files also ensures the functionality of Hot Standby as well as the interworking of TNMS and Transcend Controller (for Real-time Planning and Provisioning, etc).



Note: The FQDN (Fully Qualified Domain Name) is displayed via `hostname --fqdn`.

The examples below outline two different scenarios **with** and **without** Hot Standby. The Transcend Controller IPs are only needed if Transcend Controller is included in the network.

For networks **without** Hot Standby the *hosts* file must be configured as follows:

- TNMS Server: IP Address FQDN hostname
- Transcend Controller: IP Address FQDN hostname
- TNMS FE Server: IP Address FQDN hostname

For networks **with** Hot Standby the *hosts* files must be configured as follows:

- Primary site Server *hosts* file:
 - TNMS Primary Server: IP Address FQDN hostname
 - TNMS Secondary Server: IP Address FQDN
 - Transcend Controller Primary: IP Address FQDN hostname
 - Transcend Controller Secondary: IP Address FQDN hostname
- Secondary site Server *hosts* file:
 - TNMS Secondary Server: IP Address FQDN hostname
 - TNMS Primary Server: IP Address FQDN
 - Transcend Controller Secondary: IP Address FQDN hostname
 - Transcend Controller Primary: IP Address FQDN hostname



Tip: If there are more local IPs configured in the system, consider adding them to each *hosts* file to improve system diagnostic operations.

Host name valid characters are:

Table 10: Valid characters for host name

- (must not be at the beginning or end of the name)
1234567890
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ



NOTICE: The TNMS installer will check if the *hosts* file is correctly configured. If the server belongs to a domain, make sure FQDN matches the domain.

If no domain exists and the *hosts* file is not configured, the installation will not proceed.

4.2. Virtual Memory

The system configuration file (*/etc/sysctl.conf*) must be updated with the following settings:

```
vm.swappiness = 1
```



NOTICE: The following parameters are not applicable to the TNMS Mediation Machine in a Special Deployment scenario:

```
vm.min_free_kbytes = 1048576
```

```
vm.dirty_ratio = 15
```

```
vm.dirty_background_ratio = 3
```

The second parameter configures the system to only use swapping as a last resource, that is, the system will use paging when needed. The remaining parameters are RHEL advised parameters.

4.3. FTP Configuration



NOTICE: Perform this configuration only if the network contains 5500 DCC NEs or Embargo NE versions that support only FTP and need to be managed by TNMS.

To configure the FTP, do as follows:

Step 1. Login as *root*.

Step 2. Edit */etc/vsftpd/vsftpd.conf*:

Step 2a. Set the following property to:

```
anonymous_enable=NO
```

or add the line above if this property is not in the file. After adding this line, save and exit.

Step 2b. (Only required if IPv6 is disabled)

- Set the following property to: `listen=YES`

or add the line above if this property is not in the file.

- Comment out the following property: `# listen_ipv6=YES`

Step 2c. After editing the file, save and exit.

Step 3. (Only required if SELinux is enabled)

Allow the FTP daemon to access the filesystem:

Linux 8.X and Linux 9.X:

```
# setsebool -P ftpd_full_access on
```

Step 4. Restart the `vsftpd` service:

Linux 8.X and Linux 9.X:

```
# systemctl restart vsftpd
```

4.4. Network Time Protocol

4.4.1. How NTP works

TNMS uses Network Time Protocol (NTP) and dedicated time servers to synchronize and manage NEs. The servers have a hierarchical, Client-Server relationship between them. [Table 11: NTP hierarchy of management components \(p. 31\)](#) shows an example of such a hierarchy with four Clients. Network components on stratum level n use the stratum level $n-1$ component as time server.

Table 11: NTP hierarchy of management components

Network component	Type	Time server	Stratum
Top Level Time Server	GPS clock		0
Server	LINUX server	Top Level Time Server	1
Client 1, Client 2, Client 3, Client 4	Windows PC	Server	2

If the NE is a Gateway NE (GNE), the respective time server is the TNMS Server. If the NE is reached via embedded channels, there will be an NTP chain between these NEs. [Table 12: NTP hierarchy of NEs \(p. 31\)](#) shows the typical configuration, starting at the Server, serving time to the GNE, assuming four elements in the NE chain. The numbers in brackets indicate the stratum level in case GNE 1 fails.

Table 12: NTP hierarchy of NEs

Network component	Type	Time server	Stratum
Server	LINUX server	Top Level Time Server	1
NE 1 (GNE)	LINUX	Server (NE 2)	2 (5)
NE 2	LINUX	NE 1 (NE 3)	3 (4)
NE 3	LINUX	NE 2 (NE 4)	4 (3)
NE 4 (GNE)	LINUX	NE 3 (Server)	5 (2)

Each network component requires a specific installation and configuration. In an NTP network, each node is identified by its IP address or computer name, and holds a configuration file (`ntp.conf`) referring to its time servers by IP address.



Note: Although NEs may also have an *ntp.conf* file, EMs handle their NTP management. For information on the NEs' NTP configuration see NE-specific documentation.

4.4.2. Configuring NTP

Configure the Server as follows:

Step 1. Edit configuration file */etc/chrony.conf*

Issue:

```
# vi /etc/chrony.conf
```

Step 2. Add the NTP server address and other NTP configurations to the *chrony.conf* file:

```
'server aaa.bbb.ccc.ddd'
```

Where "aaa.bbb.ccc.ddd" is the external NTP Server's IP.

It is recommended to set two external NTP Servers, a Primary and a Backup.

Step 3. Enable the NTP service.

Issue:

```
# systemctl enable --now chronyd
```

Step 4. Check the NTP service status and start it if necessary.

Issue:

```
# systemctl is-active chronyd
```

```
# systemctl start chronyd
```

4.5. Secure Shell/SSH File Transfer Protocol

It is necessary to configure the SSH File Transfer Protocol (SFTP) settings for correct use in TNMS.



Note: Networks that include 7100 Nano or 7100 OTS NEs are not compatible with SFTP in RHEL 9. Refer to the *TNMS Troubleshooting Manual > SFTP Operations Failing in RHEL 9 for 7100 Nano or 7100 OTS NEs* chapter to configure an external SFTP server.

4.5.1. Recommend SFTP Settings:

Step 1. Log in as *root*.

Step 2. Edit the SSH daemon configuration file.

Issue:

```
# cd /etc/ssh
```

```
# vi sshd_config
```

Step 3. Set the following:

```
MaxStartups 150
```

```
MaxSessions 100
```

```
TCPKeepAlive yes
```

```
ClientAliveInterval 60
```

```
ClientAliveCountMax 5
```

Step 4. Make changes effective by restarting the daemon.

Issue:

```
# systemctl restart sshd
```

4.5.2. Enable SHA-1

SHA-1 usage for signatures is restricted in the default system-wide cryptographic policy in RHEL 9. For networks that include mTera and hiT 7300 NEs with the TNMS Server in RHEL 9, SHA-1 must be enabled:

Step 1. Login as *root*

Step 2. Enable SHA-1.

Issue:

```
update-crypto-policies --set DEFAULT:SHA1
```

Step 3. Reboot the system.

Issue:

```
reboot
```

4.5.3. Enable DSA Host Keys

Only required for networks that include 7100 Nano or 7100 OTS NEs and the TNMS Server is in RHEL 8.

OpenSSH packaged in Linux disables DSA host keys by default. The keys must be regenerated and manually enabled in TNMS Servers / TNMS Mediations running in Linux RHEL 8 that manage 7100 Nano or OTS NEs. To regenerate the keys, do as follows.

Step 1. Login as *root*.

Step 2. To create new DSA keys run:

```
# ssh-keygen -t dsa -f /etc/ssh/ssh_host_dsa_key
```



Note: The passphrase MUST be empty.

Step 3. Edit */etc/ssh/sshd_config* and add or uncomment the following lines:

```
HostKey /etc/ssh/ssh_host_dsa_key  
PubkeyAcceptedKeyTypes+=ssh-dss  
HostkeyAlgorithms+=ssh-dss
```

Step 4. *Optional step, only required if SELinux is being used.*

Issue:

```
# restorecon -v /etc/ssh/*
```

Step 5. Edit */etc/sysconfig/sshd* and add or uncomment the following line:

```
CRYPTO_POLICY=
```

Step 6. Restart ssh service.

Issue

```
# systemctl restart sshd
```

4.6. Proxy Settings

If the environment variable `http_proxy` is configured in the operating system, make sure it is defined for all users using this syntax:

```
http_proxy=http://proxy.domain:3128
```

where `proxy.domain` is the name of the proxy server with the network domain.

4.7. Required Configuration Values

This chapter describes the specific configurations set at installation that must not be modified to enable the correct functioning of TNMS.

4.7.1. Umask Configuration

The root user must have the umask default value set to `0022`.

 **Note:** Modifying the umask default value is not supported.

To verify the umask default value:

Step 1. Log in as root

Step 2. Execute the below command:

Issue:

```
# umask
```

The command output should be: `0022`

4.7.2. No `noexec` in `/tmp`

During software installation and priority update installation there should be no `noexec` attribute set for `/tmp` partition.

To achieve no `noexec` in `/tmp` partition:

Step 1. Execute the below command to check the partition attributes.

```
cat /etc/fstab | grep /tmp
```

Step 2. Remove `noexec` attribute and remount the partition if set.

After installation the partition attributes can be restored to original values.

4.7.3. Kernel Parameters

The kernel parameters `kernel.shmall` and `kernel.shmmax` should be their initial values, or, above the minimum values recommended by Oracle, which are:

- `shmmax` should be half the physical memory size, measured in bytes.

To verify `shmmax` value, execute:

```
cat /proc/sys/kernel/shmmax
```

- `shmall` should be greater than or equal to value of `shmmax`, measured in pages.

To verify `shmall` value, execute:

```
cat /proc/sys/kernel/shmall
```

4.8. Other Configurations

4.8.1. Extended File System Attributes

This is only applicable to ext4 filesystem:

Check that the extended attributes of the file system are enabled.

To do so, check the properties of the device where the file system is mounted:

Step 1. Find the device:

Issue:

```
# df <Product_Installation_Folder>
```

Output:

```
Filesystem 1K-blocks Used Available Use% Mounted on
/dev/mapper/rootvol-root 72117368 45155488 23275480 66% /
```

Step 2. Check the properties of the device:

Issue:

```
# /sbin/dumpe2fs -h /dev/mapper/rootvol-root 2>/dev/null | grep 'Default
mount options'
```

Output:

```
Default mount options: user_xattr acl
```

If the default mount option is **user_xattr** the file system has the extended attributes enabled.

4.8.2. File System ACL

This is only applicable to ext4 filesystem:

Step 1. Edit the file */etc/fstab*.

Step 2. Find each line with: *<dir> /, <dir> /oradata/ora1, <dir> /oradata/ora2, <dir> /oradata/ora3, <dir> /nokia*. For example:

Table 13: File System Mounting Options 1

<device>	<dir>	<type>	<options>	<dump>	<fsck>
/dev/sda1	/	ext4	defaults,acl	0	1

or:

Table 14: File System Mounting Options 2

<device>	<dir>	<type>	<options>	<dump>	<fsck>
UUID=e3a3a46c-4730	/	ext4	defaults,acl	0	1

Step 3. In each line referring to the mountpoint directories listed in the previous step, check if **,acl** is written after **defaults** and, if not, add it. Otherwise no change is required.

Step 4. Restart the machine to apply the permissions.

4.8.3. Swap Configuration

TNMS requires a swap configuration of at least 50% of physical memory size. When the TNMS Server and TNMS Mediation are installed in different machines (TNMS Special Deployment), TNMS mediation requires a swap configuration of 100% of the physical memory size. Check the configured swap by issuing:

```
# swapon -s
```

In the following example of a system output, size is displayed in bytes:

Filename	Type	Size	Used	Priority
/dev/sda6	partition	15938556	0	-1



Note: If no output is provided then no swap is configured in the system, to confirm use the `free` command. Refer to chapter [Disk Configuration \(p. 19\)](#) for the required swap configuration.

4.8.4. Firewall Configuration

By default, the firewall is enabled after the operating system installation. To connect TNMS Client from remote clients, the firewall must be temporarily disabled.

Issue:

```
# systemctl stop firewalld  
# systemctl disable firewalld
```

For security purposes, the firewall (i.e iptables) must be enabled. Refer to chapter Networking and Firewall Configuration in the *TNMS Administration Manual*.

5. Software Prerequisites Installation



Tip: It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)* spreadsheet.

For more information regarding this checklist refer to [TNMS Installation Checklist \(p. 12\)](#).



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

5.1. Database Software Download and Preparation

In terms of licensing, TNMS requires Oracle Database Server Enterprise Edition with the partitioning option. The database is installed and automatically configured to comply with the Center for Internet Security (CIS) security recommendations.

Extract the *V9.1 DB SW Installation (Linux)* archive (downloaded from the Customer Portal) to a new location. The default path is `/opt/oracle/oramedia`:

Step 1. Log in as *root*.

Step 2. Create the Database Software installation directory.

Issue:

```
# mkdir -p /opt/oracle/oramedia
```

Step 3. Extract the *V9.1 DB SW Installation (Linux)* archive to `/opt/oracle/oramedia/`



Note: The number of database *zip* files in the *V9.1 DB SW Installation (Linux)* archive varies from release to release. Refer to the *Release Notes* for exact files included in TNMS release.

5.2. TNMS Database Installation and Configuration

Step 1. Log in as *root*.

Step 2. Insert the TNMS software for Linux disc and mount the drive unit.

Step 3. Extract the content of TNMS software prerequisites to a directory with read permissions for all users, for example, `/install`.



NOTICE: The database software may be incorrectly installed if you:

- Run the installation script (`./installation.sh`) from within a directory without read permissions, such as `/root` or `/home`.
- Perform any changes (directory structure, files and their location) to the extracted files and folders.

Step 4. Ensure that all extracted files are readable to all users and the directories have execution permissions.

Issue:

```
# find /install/TNMS_Prerequisites/Oracle -type d -exec chmod 755 {} \;  
# find /install/TNMS_Prerequisites/Oracle -type f -exec chmod 644 {} \;  
# find /install/TNMS_Prerequisites/Oracle -name '*.sh' -exec chmod 755 {} \;  
\;
```

Step 5. Go to `<path>/TNMS_Prerequisites/Oracle/installation`.

Step 6. Issue as *root*:

```
# ./installation.sh
```

Step 7. Accept the End User License Agreement.

Step 8. Accept the configuration of Huge Pages by the Database.

Step 9. Select your configuration. by entering *S* for Small, *SP* for Small Plus, *M* for Medium or *L* for Large.

Step 10. If a Large Configuration was selected in the previous step, choose if Node Manager should be installed.

Step 11. Enter the ORADATA path, or accept the default, `/oradata`.

Step 12. If required, enter the Installer zip file, or accept the default.

This file is in `/opt/oracle/oramedia`, if the database software zip files were copied to the default location.

Step 13. If required, enter the TNMS Installer folder path, or accept the default.

Step 14. If required, enter the `TNMS.rsp` path or accept the default.

Step 15. If required, enter the template file path or accept the default.

The installer will now verify the requirements as shown in the example below:

```
Verifying requirements...  
..-Verifying system requirements...done.  
..-Verifying disk requirements...done  
..-Verifying memory requirements...done  
..-Verifying swap requirements...done  
..-Verifying hostname...done  
..-Verifying if NIS domain name is empty...done  
..-Verifying User and Group configuration...done.  
..-Verifying directories... done.
```

If any error message is displayed (for example: Error in the `/etc/hosts` file), correct it before proceeding to the next step.

As an example, an error in memory configuration is shown below:

```
..-Verifying memory requirements...Error. Check RAM memory. Total: 8 GB.  
Required: 16 GB. Error checking requirements. Are you sure you want to  
continue (YES/no)? [no] :
```

Step 16. Enter a name with 1 to 8 characters for the TNMS database, or accept the default name, `TNMS`. Valid names start with an upper case character (A-Z), followed by upper case characters (A-Z) or numbers (0-9).

Step 17. Enter the listener port or press **Enter** to use the default port [1521].

Step 18. Type the password for the default database administrator user `SYS` . Refer to [Database User Password Complexity Rules \(p. 50\)](#) section.

 **Note:** Keep the username and password for future reference.

Step 19. Type the `SYS` password again.

Step 20. Type the password for the default database administrator user `SYSTEM` with at least 6 characters and a maximum of 30.

Refer to [Database User Password Complexity Rules \(p. 50\)](#) section.

 **Note:** Keep the username and password for future reference.

Step 21. Type `SYSTEM` password again.

Afterwards the installation process installs and configures the database.

Wait until the installation process is finished. This step may take several minutes and the progress can be seen in `/tmp/oracle_installation:<timestamp>.log`.

At the end of a successful installation the following message is displayed:

Final status of the execution: Success

 **Note:** At the end of the installation, the Oracle logs are written to the following folder: `/home/oracle/ossnms_installation_log`

 **NOTICE:** The operating system user `orabackup` has been created and added to the `dba` group. This user must have no password expiration date and must not be deleted or removed from this group.

Step 22. After a successful installation of the TNMS Database, navigate to the Database Software installation folder (by default `/opt/oracle/oramedia`).

Step 23. Delete the *V9.1 DB SW Installation (Linux)* zip files and the corresponding extracted folders.

5.3. Installing Database Security Patches

Oracle publishes regular patch updates with fixes for multiple security vulnerabilities. These patches must be applied to ensure correct system behavior.

 **Note:** For a comprehensive list of the Database Security patches applicable to this release, please refer to the Database Security patches chapter in the *Release Notes Appendix*.

Before proceeding, extract the *V9.1 DB Security Patch (Linux)* archive (downloaded from the Customer Portal) to a new location (by default `/opt/oracle/oramedia`):

Step 1. Log in as `root`.

Step 2. Create the directory (by default `/opt/oracle/oramedia`), if it was not already created in [Step 2 \(p. 37\)](#).

Issue:

```
# mkdir -p /opt/oracle/oramedia
```

- Step 3.** Extract the *V9.1 DB Security Patch (Linux) archive* to */opt/oracle/oramedia/*
This archive contains the Database Security patches (*p<number>_<version>_<Linux_Version>.zip*)
and the Patch Installation *zip* file (*patch.zip*).

5.3.1. Installing Database Security Patches

- Step 1.** In the TNMS Server Machine, log in as *root*.
- Step 2.** In the *oramedia* directory (by default */opt/oracle/oramedia*), extract the *patch.zip* file and execute.
Issue:

```
# chown -R oracle:dba .  
# chmod 744 ./patch/patch.sh
```

- Step 3.** Apply the Database Security patches as Database software owner.
Issue:

```
# su - oracle  
# cd /opt/oracle/oramedia/patch (if default path was chosen)  
# ./patch.sh
```

- Step 4.** Enter the directory where the Database Security patches are located, or accept the default (*/opt/oracle/oramedia*).

i **Note:** If a database shutdown is required, the script will ask you to shutdown the application before proceeding. If applicable, type **y** to continue.

The script can be re-executed in case of failure.

A successful execution ends with the message:

Final status of the execution: Success

i **Note:** At the end of the installation, the Database logs are written to the following folder:
home/oracle/ossnms_installation_log

- Step 5.** After a successful installation of the TNMS Database Security Patches, navigate to the Database Software installation folder (by default */opt/oracle/oramedia*).

- Step 6.** Delete the following files:
- The *V9.1 DB Security Patch (Linux) archive*.
 - All Database Security patches zips (*p<number> <version> <Linux_Version>.zip*)
 - The Patch Installation *zip* (*patch.zip*)
 - All extracted archives: directories *patch* and directories *<number>*

5.3.2. Security Patches Installation Troubleshooting

i **Note:** Consult the individual patch logs for more details about the errors in the table below. The logs can be found in
/home/oracle/ossnms_installation_log/ossnms_patch_installation_<timestamp>_<timestamp>.p>.

Table 15: Security Patch Logs

Message	Cause	Solution
OPatch failed with error Code = 73 Prerequisite "CheckSystem-Space" failed. Required amount of space is not available.	Not enough free space in drive	Free the amount of space required and reapply the patch,
Error applying datapatch in [database name]	Internal datapatch issue	Restart patch installation.

6. TNMS Installation

This chapter describes the TNMS installation. **If you have a previous TNMS version installed in your system, jump to [Update TNMS \(p. 65\)](#).**

Before you install TNMS be sure to read and follow the directions below. **Failing to comply will result in a failed installation.**



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

6.1. Installation Packages

TNMS is delivered with a Full Installation Package (**Main**), as well as Legacy LCTs Package (**Legacy LCTs**).

- **Main:** includes TNMS, OEMs and LCTs.
- **Legacy LCTs:** includes legacy LCTs and their required OEMs. For a list of Legacy LCTs refer to the TNMS Release Notes, Supported NEs chapter.



Note: For more information on upgrading TNMS refer to the *TNMS Upgrade Manual*.



Tip: It is highly recommended to take note of relevant configurations using the *TNMS Installation Checklist (Linux)* spreadsheet. For more information regarding this checklist refer to [TNMS Installation Checklist \(p. 12\)](#).

6.2. TNMS Installer Integrity Validation

Before proceeding with the TNMS Installation, the Full Installation Package (**Main**) must first be verified and extracted. This must be done in each machine that the TNMS Server, TNMS Mediation or TNMS Client will be installed.

Step 1. Verify the checksum against the one provided.

Issue:

```
# sha256sum <Installer Zip File>
```

Step 2. Extract the TNMS software to a directory in the machine, now referred to as *<Product_Installer_Folder>*.



NOTICE: The *<Product_Installer_Folder>* must not be in the same directory as the *<Product_Installation_Folder>* or the *<Product_Data_Folder>*.

Step 3. If Legacy LCTs are required:

Step 3a. Verify the checksum against the one provided

Issue:

```
# sha256sum TNMS_Legacy_LCTs.zip
```

Step 3b. Extract the TNMS Legacy LCTs to the *<Product_Installer_Folder>*

Step 4. Verify that the latest available 9.1 PDTs in the Nokia Customer Portal are downloaded and stored in *<Product_Installer_Folder>\TNMS_Installer\PUs*.

6.3. Prerequisites and Validation before TNMS Installation

Ensure prerequisites are validated before installing TNMS.

 **Note:** This script is applicable only for Linux 9.X.

Step 1. Go to *<Product_Installation_Folder>/TNMS_Installer/verify_prerequisites*.

Step 2. Set execution permissions for the *verify_prerequisites_tnms.sh* script.

Issue:

```
# chmod +x verify_prerequisites_tnms.sh
# ./verify_prerequisites_tnms.sh
```

Follow the instructions. The script checks requirements and concludes with one of the following messages:

Error: Missing requirement.

Warning: Review needed before installation.

Ok: All requirements met.

 **Note:** Ensure all errors are corrected before installation.

6.4. TNMS Server and Mediation Installation

To install TNMS Server and Mediation:

Step 1. Log in as *root*.

Step 2. Configure the TNMS FE Server connection.

 **Note:** This step is only applicable if TNMS FE Servers are included in the system.

Configure the Server as follows:

Step 2a. Edit the file:

```
/opt/oracle/product/19c/dbhome_1/network/admin/sqlnet.ora
```

Step 2b. Add the IPs of the TNMS FE Server machines to the property

```
TCP.INVITED_NODES.
```

For example:

```
TCP.INVITED_NODES=(127.0.0.1, loghost)
```

should be changed to

```
TCP.INVITED_NODES=(127.0.0.1, loghost, 10.79.32.160,10.46.84.68)
```

where 10.79.32.160 and 10.46.84.68 are the TNMS FE machine.

Step 2c. Reload the database listener.

Issue:

```
# su - oracle
lsnrctl stop lisner
lsnrctl start lisner
exit
```

Step 3. Go to *<Product_Installer_Folder>*, created during the [TNMS Installer Integrity Validation \(p. 42\)](#) and manually assign *Execution* permissions.

Issue:

```
# cd <path>/TNMS_Installer
# chmod 744 ./TNMS.bin
```

Step 4. Start the installation wizard.

Run:

```
# ./TNMS.bin
```

The installation wizard opens in **Introduction** and the complete list of installation steps is displayed on the left pane.

Step 5. Read the **License Agreement** and select *I accept the terms of the License Agreement*.

Step 6. In **Installation Package**, click **TNMS Server and Mediation**.

Step 7. In **Server Preparation: Transcend Controller**:



Note: This step is only applicable if Transcend Controller is included in the system

Step 7a. Click **Enable Connection to Transcend Controller (TC)** to configure the following parameters:

- TC IP: Transcend Controller IP address.
- TC OIF Port: Transcend Controller OIF Port (Default is: 12443).
- TC Gateway Port: Transcend Controller Gateway Port (Default is 12351).



NOTICE: This is needed for the correct interworking of TNMS and Transcend Controller functionalities (for example Hot Standby, Real-time Planning and Provisioning, etc). Before proceeding, ensure that the Transcend Controller Server IP and FQDN have been configured, refer to chapter [System Hosts \(p. 29\)](#).

Step 8. In **Server Preparation: Hardware Configuration** select your hardware configuration: **Small, Small Plus, Medium** or **Large** (see [Hardware Requirements \(p. 13\)](#)).

Step 9. In **Server Preparation: Customization**, select items to customize:

- For each checked item the respective customization step is displayed. Either accept or change the default values.
- Unchecked items will automatically apply defaults values to the installation.

Step 9a. Users and groups checked
In Users and Groups enter:

- TNMS **User** Name (default is *tnms*).
Checking **Create** enables *User ID*
- TNMS **Group** (default is *tnms*).

Checking **Create** enables *Group ID*

- **SFTP User Name** (default is *tnms_sftp*).
Checking **Create** enables *User ID*
- **Database User Name** (database owner name. Default is *oracle*).



Note: The *User ID* and *Group ID* values must be numeric.

- **DBA Group Name** (default is *dba*).



NOTICE: If the warning message **Users or usergroup being created already exist** is displayed, before proceeding delete the impacted user or user group.

- i. Login as root.
- ii. Delete applicable users or user groups.

If default user and user group names are used, issue the applicable command:

- `userdel tnms`
- `userdel tnms_sftp`
- `groupdel tnms`

Step 9b. **Deployment Directories** checked

In Deployment Directories enter the path of the following folders (default paths are provided):

- **TNMS Installation Directory**

If you choose a directory other than the default, the path cannot contain spaces.



Note: Throughout the customer documentation this folder will be referred to as <Product_Installation_Folder>. Take note of the location of this folder for future reference.

- **TNMS Data Directory**

If you choose a directory other than the default, the path cannot contain spaces.



Note: Throughout the customer documentation this folder will be referred to as <Product_Data_Folder>. Take note of the location of this folder for future reference.

- **Database Installation Directory** (default is */opt/oracle*)

- **Database Data Directory**

Enter the path you defined in 5.1.1 Installation and Configuration (default is */oradata*). Backups will be stored under subfolders of this folder.

Step 10. *Only displayed if more than one IP is configured.*

In **Server Preparation: Connection Configuration**, enter the required IP for:

- **Client Access:** The IP Address through which the TNMS server communicates with TNMS Clients.
- **Server Backend Access:** The IP Address through which the TNMS server communicates with other connected systems.

In **Special Deployments:**

- **TNMS Server installation:** this is the IP Address through which the TNMS Server communicates with TNMS Mediation.
- **TNMS Mediation installation:** this is the IP Address through which TNMS Mediation communicates with the NE Network.

Step 11. In **Server Preparation: Database** select:

- **New:** creates a clean database
- **Migration:** migrates an existing database.
Use this option to add managers and NBIs to your database. The components already present in the database will be grayed out throughout the installation wizard. Some parameters will also not be displayed as they are defined in the database.



Note: Please refer to the *TNMS Upgrade Manual* for the supported migration paths.

Step 12. In the **Server Preparation: Database Connection** enter the required information for the **TNMS Database** connection:

- **Database IP Address:** IP assigned to the database to communicate with TNMS.
- **Localhost:** Hostname of the local computer running TNMS (Default is *127.0.0.1*)
- **Database Port:** Port assign to the database to communicate with TNMS (Default is *1521*).
- **Database User Name:** Name of the database user (Default is *tnmsdba*).



NOTICE: Special characters are not allowed when selecting a different user name for the Database.



NOTICE: If a standby server is deployed in the network, the same user name and password must be used for both the Primary and Secondary Servers. It is recommended that the same user name and password are used in all server machine installations in the network, ensuring database restoration in any machine. If a standby server is not deployed in the network, unique user name and passwords can be used for each server machine if required for security reasons. Recording and storing all server user names and passwords in the Installation Checklist (refer to chapter [TNMS Installation Checklist \(p. 12\)](#)) is highly recommended to ensure that backup and restore can be performed when required in future.

- **Database User Password:** Password for the database user that complies with the **Password Complexity Rules**.



Tip: Keep the user name and password for future reference, for example, in the [TNMS Installation Checklist \(p. 12\)](#).

- **Re-enter Database User Password:** Re-enter the password.
- **Database Name (SID):** Name of the database that has been created previously (DB instance), which, by default, is TNMS (Refer [5.1 Database Software Download and Preparation \(p. 37\)](#))
- **Database User 'sys' Password:** Password defined in [5.1 Database Software Download and Preparation \(p. 37\)](#) for the default database administrator user SYS.
- **Home Directory:** Database home directory (Default is /opt/oracle/product/19c/dbhome_1).

Step 13. In Server Preparation: Advisory Message

- Enable or disable an advisory message before login.
- If enabled, write the desired advisory message in the box.

This message will pop-up before a user's login.

Step 14. In Components:TNMS select which managers to install.

Selecting **Embedded DNA** will install eDNA Server. *Applicable only for machines with a Medium or Large hardware configuration.*



NOTICE: Node Manager and eDNA cannot be installed in the same machine



NOTICE: The TNMS Server supports:

- A maximum of fifteen (15) eDNA desktop clients in any network. If there are more than fifteen (15) eDNA desktop clients, TNMS FE Servers are required. TNMS FE Servers support a maximum of fifteen (15) eDNA clients each. If TNMS FE Servers are included in the network, eDNA desktop clients connect to the TNMS FE Servers in a round robin fashion. For more information refer to [TNMS Special Deployment \(p. 50\)](#).
- A maximum of two (2) simultaneous eDNA webclient sessions in a Large or Medium configuration. Support for more eDNA webclients requires TNMS FE Server machines. Each TNMS FE Server machine supports up to 15 eDNA webclients.

Each TNMS component in this step requires a license, except Node Manager and eDNA.

Step 15. In Components: TNMS FE Servers enter a list of hostnames, fully qualified, and respective IPs in the following format:

<hostname>.<domainname>=<IP>

For example: funchal.nokia.com=10.1.1.2

Step 16. Only displayed if you selected Embedded DNA in step 13,

In **Components: DNA FE Servers** enter a list of hostnames, fully qualified, and respective IPs in the following format:

FE_<hostname>.<domainname>_BIND_IP=<IP>

For example: FE_funchal.nokia.com_BIND_IP=10.1.1.2

Step 17. In **Components:Northbound Interfaces** select the North Bound Interfaces to install.

- SNMP
- TMF CORBA NBI

By default the TMF CORBA NBI interface uses the TMF 814 MESH topology.

Check Legacy Mode only if you are migrating from TNMS Core, with TCOA integration, and need to maintain the TMF 814 SINGLETON topology.



Note: If you install TMF CORBA NBI and if TNMS Server is running in a machine with two or more network interfaces you must, after the TNMS Server installation, perform the configuration described in 7.4 [Configuring NTI Notification Service Address \(p. 60\)](#).

- TRANSCEND REST NBI



Note: Refer to chapter Northbound Interfaces, in the *TNMS Administration Manual*, for more information.

Step 18. In the **Components: Network Elements** select the NEs to install.

Step 19. *Only displayed if you selected to migrate your database.*

In **Pre-migration Backup** your current database to avoid data loss if any issue occurs during the migration.

Step 20. If you selected to install Node Manager, enter the **FTP/SFTP Settings** for Node Manager to communicate with the 5500 NEs. Skip this step in case of other NE types.
These FTP/SFTP settings are the ones that TNMS will also use.



Note: These settings must match those configured during the [Setting Up the Password for TNMS, SFTP and FTP User \(p. 62\)](#) procedure. The [TNMS Installation Checklist \(p. 12\)](#) provides fields for the recording of these settings.

Step 21. *Only displayed if you chose to install Node Manager in a Large configuration.* In **Server Preparation: Database Connection** enter the required information for the **Node Manager** Database connection:

- **Database IP Address:** default is *127.0.0.1*, shown if *localhost* is checked.
- **Database Port:** default is *1521*.
- **Database User Name:** the user that owns the scheme of the database to be created. This user name should be the TNMS database username defined in the previous step plus “_NM”, such as in *tnmsdba_NM*.



NOTICE: For networks with a Standby Server, the username and password for both the Primary and Secondary Servers must be the same. The same approach is also recommended for all TNMS database installations to ensure that the database is restorable in any machine. The [TNMS Installation Checklist \(p. 12\)](#) provides fields for the recording of usernames and passwords.

- **Database User Password:** password for the Node Manager database user that complies with the Password Complexity Rules.



Note: Keep the username and password for future reference.

- **Re-enter Database User Password:** re-enter the password.
- **Database Name (SID):** the name of the Node Manager database which is NMDB.



NOTICE: The Node Manager database name cannot be the same as the TNMS database name.

- **Database User 'sys' Password:** Password defined in 5.1.1 Installation and Configuration for the default database administrator user SYS.

Step 22. A summary of the installation settings is displayed in the Pre-Installation Summary step. If PDTs were stored in the `<Product_Installer_Folder>\TNMS_Installer\PUs` folder, the PDT numbers will be displayed for verification. Confirm all configurations are correct and click Install.



NOTICE: If the firewall is enabled, a warning message is displayed stating Enabled Firewall detected.

For the full list of all the ports that need to be opened in the firewall for the correct functioning of TNMS refer to the Communication Matrix.xls (`<Product_Installation_Folder>\client\help`). To obtain a direct correlation between a previously installed version with the current ports, see the Instructions worksheet for more information.

Step 23. *Only displayed if you selected to migrate your database.*
In **Post-migration Backup** your database after migration.

Step 24. The results of the installation are presented in **Installation Results**.
Click **Done** to finish the installation.

Step 25. The TNMS environment variables must be updated post installation.
Execute:

```
#. /etc/profile.d/ossnms.sh
```

Step 26. After a successful installation of TNMS Server and Mediation, navigate to the `<Product_Installer_Folder>`.

Step 27. Delete the TNMS Software zip files and the `<Product_Installer_Folder>`.

Step 28. Consult the Technical Notes from all applicable Priority Updates or Workarounds for updated recommendations or additional requirements.



Note: The TNMS Server machine hostname should be resolved through the DNS Servers on the Client side. Update of the NGINX CA and host certificate is also required, see in 2.4.2 chapter of the *TNMS Administration Manual*.

6.4.1. Password Complexity Rules

The passwords are validated by the system according to the rules below:

- The password must have between 8 - 32 valid characters (see [Table 16: Valid Characters \(p. 50\)](#)).
- The password must not contain the username, the reversed username nor a circular shifted version of the username.
- The password must not contain sequences of three or more characters of the user name.
- The password must not contain more than three repeated characters of the same type, either lower or upper-case, for example *aAaA*.
- The password must not contain more than three consecutive characters in ascending or descending order, either lower or upper-case, for example *aBcD*.
- The password must not contain a sequence of two or more repeated characters, for example *a12b12*.
- The password must not begin nor end with a space.
- The password must include at least three of the following four specifications: one lower case alpha character, one upper case alpha character, one numeric character and one special character.

6.4.2. Database User Password Complexity Rules

The password must have between 6 - 30 valid characters (see [Table 16: Valid Characters \(p. 50\)](#)).

Table 16: Valid Characters

+ - : { }
1 2 3 4 5 6 7 8 9 0
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

6.5. TNMS Client Installation

For the TNMS Client Installation procedure please refer to chapter TNMS Client Installation in the *TNMS Installation Manual Windows*.

6.6. TNMS Special Deployment

The procedures below describe how to install a TNMS Server and TNMS Mediations in different machines. However, if you intend to proceed with a special deployment, first contact Nokia Technical Support or Product Management, so they can advise you on finding the best solution for your business.

Nokia recommends that you install one TNMS mediator per TNMS Mediation machine, however the Node Manager mediator can be installed in a TNMS Mediation machine also.

Installation Sequence

The TNMS Server must be installed first, followed by TNMS and Node Manager Mediation(s) afterwards.

6.6.1. TNMS Server Installation

Hardware requirements for TNMS Server installation

The hardware requirements for the installation of the TNMS Server without Mediations are the same as described in [Table 3: Hardware requirements for installations of TNMS \(p. 14\)](#)

TNMS Server Installation Procedure

To install TNMS Server without the Mediation or Transcend eDNA FE Servers separately, proceed as follows:

Step 1. Log in as `root`.

Step 2. Configure the Database Server as follows:

Step 2a. Edit the file:

`/opt/oracle/product/19c/dbhome_1/network/admin/sqlnet.ora`

Step 2a. Add the IPs of the Mediation machines (where Node Manager will be installed) and of the eDNA FE machines to the property `TCP.INVITED_NODES`. For example:

`TCP.INVITED_NODES=(127.0.0.1, loghost)`

should be changed to:

`TCP.INVITED_NODES=(127.0.0.1, loghost, 10.79.32.160,10.46.84.68)`

where 10.79.32.160 is the Mediation machine that will have Node Manager installed and 10.46.84.68 is the eDNA FE machine.

Step 2b. Reload the database listener.

Issue:

```
# su - oracle
lsnrctl stop lisner
lsnrctl start lisner
exit
```

Step 3. Follow the procedure described in 6.1 TNMS Server and Mediation Installation. Skip the any steps that are not displayed in the wizard because they do not apply to the TNMS Server installation.

Step 4. After a successful installation of TNMS Server and Mediation, navigate to the `<Product_Installer_Folder>`.

Step 5. Delete the TNMS Software zip files and the `<Product_Installer_Folder>`.



NOTICE: If the mandatory installation sequence was not followed and you have installed the TNMS Server *after* the TNMS Mediations with Node Manager, then you must stop and start all Node Manager services in the following sequence:

- Stop all Node Manager services in the TNMS Server and TNMS Mediations.
- Start the Node Manager services in the TNMS Mediations and wait until all those services are *running*.
- Start the Node Manager services in the TNMS Server.

This sequence also applies to any other issue that requires the Node Manager services to be stopped are started.

6.6.2. TNMS Mediation Installation

Hardware requirements for TNMS Mediation installation

The table below provides an overview of the hardware recommendations for TNMS Mediation.

Table 17: Minimum hardware requirements for TNMS Mediation installations.

CPU	16 Threads, 2,8 GHz
RAM	8 GB
HDD	100 Gb

These hardware requirements refer to virtual machines. If you use physical machines the hardware requirements must be compatible with the ones defined for virtual machines.

Supported Mediators

- Generic Mediator (NEs of EM-GM type)
- MVM (NEs of EM-MVM and EM-NMS type)
- Node Manager



Note: For Node Manager to work you must install both the MVM mediator and Node Manager. All NEs installed in the TNMS Mediations must also be installed in the TNMS Server.

TNMS Mediation Installation Procedure

To install TNMS Mediation only, proceed as follows:

Step 1. Log in as *root*.

Step 2. As *root*, edit the file */etc/sysctl.conf*.

Step 2a. Add the following lines to the end of the file or change the values if the lines already exist:

```
net.ipv4.neigh.default.gc_thresh1 = 2048
net.ipv4.neigh.default.gc_thresh2 = 4096
net.ipv4.neigh.default.gc_thresh3 = 8192
```

Step 2b. Save the file and execute:

```
# sysctl -p
```

Step 3. Extract the TNMS software to a directory in your machine.

Step 4. Go to the *<Product_Installer_Folder>*, within the directory where the TNMS software files were extracted to and manually assign Execution permissions by running the following commands:

```
# cd <path>/TNMS_Installer
# chmod 744 ./TNMS.bin
```

Step 5. Start the installation wizard.

Run:

```
# ./TNMS.bin
```

The installation wizard opens in **Introduction** and the complete list of installation steps is displayed on the left pane.

Step 6. Read the **License Agreement** and select *I accept the terms of the License Agreement*.

Step 7. In **Installation Package** click **TNMS Mediation**.

Step 8. In **Server Preparation: Customization**, select items to customize:

- For each checked item the respective customization step is displayed. Either accept or change the default values.
- Unchecked items will automatically apply defaults values to the installation.



NOTICE: If DB Connection is unchecked, a randomly generated password will be assigned. This password can be changed later, by running the password script described in *TNMS Administration Manual* 4.10.2 Changing TNMS Database Password.

Step 8a. Users and groups checked

In **Users and Groups** enter:

- **TNMS User Name** (default is *tnms*).
Checking **Create** enables *User ID*
- **TNMS Group** (default is *tnms*).
Checking **Create** enables *Group ID*
- **SFTP User Name** (default is *tnms_sftp*).
Checking **Create** enables *User ID*

Click **Next**.

Step 8b. Deployment Directories checked

In **Deployment Directories** enter the path of the following folders (default paths are provided):

- **TNMS Installation Directory**
If you choose a directory other than the default, the path cannot contain spaces.



Note: Throughout the customer documentation this folder will be referred to as *<Product_Installation_Folder>*. Take note of the location of this folder for future reference.

- **TNMS Data Directory**
If you choose a directory other than the default, the path cannot contain spaces.



Note: Throughout the customer documentation this folder will be referred to as *<Product_Data_Folder>*. Take note of the location of this folder for future reference.

- **Database Installation Directory** (default is */opt/oracle*)
- **Database Data Directory**
Enter the path you defined in 5.1.1 Installation and Configuration (default is */oradata*). Backups will be stored under subfolders of this folder.

Step 9. In **Server Preparation: Connection Configuration** enter the **IP Address** to which the TNMS Server and Network Elements you are installing will connect to.

Step 10. In **Components: NM Mediators** select which Node Manager Configuration Managers to install. These refer to the type of NEs Node Manager will be managing.



Note: It is generally recommended that you install one mediator per Mediation machine. However, installing both MVM and Node Manager in the same machine is **mandatory** for Node Manager to work in this scenario.

Step 11. In **Components: Network Elements** select the NEs to install.



Note: The TNMS mediators to be installed are automatically selected according to the NEs selected in this step.

Step 12. In **Connection Configuration: Server** enter the **IP Address** of the TNMS Server the TNMS Mediation you are installing will connect to.

Step 13. *Only displayed if you chose to install Node Manager.*

In **FTP/SFTP Configuration** enter the settings Node Manager will use to communicate with NEs. These FTP/SFTP settings are the same as the ones that TNMS will also use.

 **NOTICE:** These settings must match those that you will configure later on, as described in 7.4 SFTP and FTP Configuration. The *TNMS Installation Checklist* provides fields for the recording of these settings.

Step 14. *Only displayed if you chose to install Node Manager.*

In the **Server Preparation: Database Connection** enter the required information for the **TNMS Database** connection:

- **Database IP Address:** TNMS database IP address.
- **Database Port:** Database server port number (Default is 1521).
- **Database User Name:** TNMS database user name created in the TNMS Server installation.
- **Database User Password:** password of the TNMS database user created in the TNMS Server installation.
- **Database Name (SID):** name of the TNMS database you entered in the TNMS Server installation (DB instance).
- **Database User 'sys' Password:** Password defined in [TNMS Database Installation and Configuration \(p. 37\)](#) for the default database administrator user SYS.

Step 15. *Only displayed if you chose to install Node Manager in a Large configuration.* In **Server Preparation: Database Connection** enter the required information for the Node Manager Database connection:

- **Database IP Address:** TNMS database IP address.
- **Database Port:** Database server port number. The default value is 1521.
- **Database User Name:** Node Manager database username created in the TNMS Server installation.
- **Database User Password:** password of the Node Manager database user created in the TNMS Server installation.
- **Database Name (SID):** name of the Node Manager database you entered in the TNMS Server installation (DB instance).
- **Database User 'sys' Password:** Password defined in 5.1.1 Installation and Configuration for the default database administrator user SYS.

Step 16. A summary of the installation settings is provided in the **Pre-Installation Summary** step. If PDTs were stored in the <Product_Installer_Folder>\TNMS_Installer\PUs folder, the PDT numbers will be displayed for verification. Confirm all configurations are correct and click **Install**.

Step 17. The results of the installation are presented in the **Installation Results** step. Click **Done** to finish the installation.

Step 18. The TNMS environment variables must be updated post installation.
Execute:

```
#. /etc/profile.d/ossnms.sh
```

Step 19. After a successful installation of TNMS Mediation, navigate to the *<Product_Installer_Folder>*.

Step 20. Delete the TNMS Software *zip* files and the *<Product_Installer_Folder>*.

Step 21. Consult the Technical Notes from all applicable Priority Updates or Workarounds for updated recommendations or additional requirements.

6.7. TNMS Frontend Server Installation

To install eDNA Frontend UI Server (TNMS FE Server):

 **NOTICE:** Every TNMS FE Server supports up to 15 clients, desktop or web clients.

Step 1. Log in as *root*.

Step 2. Configure the TNMS hosts file to include the TNMS FE Servers, refer to [System Hosts \(p. 29\)](#) for instructions.

Step 3. Go to *<Product_Installer_Folder>*, created during the [TNMS Installer Integrity Validation \(p. 42\)](#) and manually assign *Execution* permissions.
Issue:

```
# cd <path>/TNMS_Installer #  
chmod 744 ./TNMS.bin
```

Step 4. Start the installation wizard.
Run:

```
# ./TNMS.bin
```

The installation wizard opens in **Introduction** and the complete list of installation steps is displayed on the left pane.

Step 5. Read the **License Agreement** and select *I accept the terms of the License Agreement*.

Step 6. In **Installation Package**, click **TNMS Frontend Server**.

Step 7. In **Server Preparation: Customization**, select items to customize:

- For each checked item the respective customization step is displayed. Either accept or change the default values.
- Unchecked items will automatically apply defaults values to the installation.

- **Users and groups checked:**

In **Users and Groups** enter:

- **TNMS User Name** (default is *tnms*).
Checking **Create** enables *User ID*
- **TNMS Group** (default is *tnms*).
Checking **Create** enables *Group ID*
- **SFTP User Name** (default is *tnms_sftp*).
Checking **Create** enables *User ID*

- **Deployment Directories** checked:

In **Deployment Directories** enter the path of the following folders (default paths are provided):

– **TNMS Installation Directory**

If you choose a directory other than the default, the path cannot contain spaces.



Note: Throughout the customer documentation this folder will be referred to as *<Product_Installation_Folder>*. Take note of the location of this folder for future reference.

– **TNMS Data Directory**

Make sure that the TNMS data folder is empty. If not, backup and remove the data or select a different folder.



Note: Throughout the customer documentation this folder will be referred to as *<Product_Data_Folder>*. Take note of the location of this folder for future reference.

Step 8. In **Components: TNMS FE Server**, select the required component:

To install a TNMS Frontend Server, select **DNA Client Service + DNA FE Server**. To install only the legacy DNA FE Server functionality, select **DNA FE Server**.

Step 9. In **Server Preparation**, enter the TNMS Server IP.

Step 10. A summary of the installation settings is displayed in the **Pre-Installation Summary** step. If PDTs were stored in the *<Product_Installer_Folder>\TNMS_Installer\PUs* folder, the PDT numbers will be displayed for verification. Confirm all configurations are correct and click Install.

Step 11. The TNMS environment variables must be updated post installation.

Execute:

```
#. /etc/profile.d/ossnms.sh
```

Step 12. Only applicable for all TNMS FE Servers that were listed during the TNMS Server installation step **Components: DNA FE Servers**.

Copy eDNA keystore and truststore to the TNMS FE Server.

Step 12a. In the TNMS Server machine, open the Administration Console

Issue:

```
# cd <Product_Installation_Folder>/server/DNA/EMS/conf/ssl
# scp DNAServer.keystore Truststore.truststore
root@<TNMS_DNA_Frontend_Server_IP>:
<TNMS_DNA_Frontend_Server_Product_Installation_Folder>/server/DNA/EMS/conf/ssl
```

Step 12b. In the TNMS FE Server machine, open the Administration Console

Issue:

```
# scs-service-restart
```

Step 13. Only applicable when adding TNMS FE Servers to a network with TNMS Server and eDNA already installed and running.



Note: Only proceed with the following procedure once all additional TNMS FE Servers have been installed as per steps 1 – 11 above.

Step 13a. Log in as *root* to the TNMS Server machine.

Step 13b. Add new FE server details in `ems.conf` in eDNA Server

Issue:

```
# cd <Product_Installation_Folder>/server/DNA/EMS/conf
# vi ems.conf
```

Add new entries directly above the line `MANUALMODE_2_SERVICES_ALLOWED=true`

`FE_<hostname>.<domain name>_BIND_IP=<IP>`

Step 13c. Execute `reconfigure_dna_cert.sh` in the TNMS server.

Issue:

```
# cd <Product_Installation_Folder>/server/DNA/install
# ./reconfigure_dna_cert.sh
```

Step 13d. Edit the `product.conf` file in the TNMS server to add the FE Server IPs.

Issue:

```
#cd <Product_Installation_Folder>/system/configuration
# vi product.conf
```

Add new entries

```
# DNA_FE_IP_LIST=<TNMS_DNA_FE_SERVER_IP_#>, <
TNMS_DNA_FE_SERVER_IP_#>
```

Step 13e. Configure the additional TNMS FE Servers to access the TNMS Database:

i. Edit the file:

```
/opt/oracle/product/19c/dbhome_1/network/admin/sqlnet.ora
```

ii. Add the IPs of the new TNMS FE Server machines to the property

`TCP.INVITED_NODES.`

For example:

```
TCP.INVITED_NODES=(127.0.0.1, loghost)
```

should be changed to

```
TCP.INVITED_NODES=(127.0.0.1, loghost,
10.79.32.160,10.46.84.68)
```

where 10.79.32.160 and 10.46.84.68 are the new FE machines.

iii. Reload the database listener.

Issue:

```
# su - oracle
lsnrctl stop listener
lsnrctl start listener
exit
```

Step 13f. Edit file: `$BICNET_CONFIG_DIR/fe_servers.properties` and add the new TNMS FE Server host.

For example:

```
<FE Server1 FQDN>=<FE Server IP>
...
<New FE Server FQDN>=<New FE Server IP>
```

Step 13g. Restart TNMS Server services.

Issue:

```
# scs-service-restart
```

Step 13h. Copy the `DNAserver.keystore` and `Truststore.truststore` from the TNMS Server to all TNMS FE Servers in the network.

Issue:

```
# cd <Product_Installation_Folder>/server/DNA/EMS/conf/ssl
# scp DNAserver.keystore Truststore.truststore
root@<TNMS_DNA_Frontend_Server_IP>:
<TNMS_DNA_Frontend_Server_Product_Installation_Folder>/server/DNA/EMS/co
nf/ssl
```

Step 13i. Restart all TNMS FE Servers in the network.
Log in as *root* to each TNMS FE Server

Issue:

```
# scs-service-restart
```



Note: The TNMS FE Server machines should be resolved through the DNS Servers used on the browser side. Update of the CA and host certificate is also required, see chapter 2.4.8 of the *TNMS Administration Manual*.

7. Final System Configuration

 **Note:** It is highly recommended that system security hardening be carried out before starting TNMS in a production environment, refer to *TNMS Administration Manual, Linux Security Hardening* chapter for more information.

The first installation of TNMS has a trial license of 90 days. During this period you have permissions to access all TNMS features. After the 90 days expire you need to acquire and install license keys.

For the below post-installation configuration procedures (that can be performed at any time), refer to the *TNMS Administration Manual*:

- Login (Chapter *TNMS Login*)
- Single Sign-on (Chapter *Single Sign-on Configuration*)
- Standby Server (Chapter *Standby Server*)

 **Note:** Optical Manager licenses require a TNMS Server restart after importing.

 **Note:** TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

7.1. Configuring NGINX Service

NGINX service default port (8444) is configurable, in case any other software installed is already using port 8444. Follow the procedures below to configure the NGINX port.

7.1.1. Configure NGINX in TNMS Mediation

Step 1. Stop the TNMS nginx service.

Issue:

```
#scs-service-stop nginx
```

Step 2. Open the file:

```
<Product_Installation_Folder>/system/oper/nginx/conf/nginx.conf
```

Step 3. Edit the file, replacing the port number in listen 0.0.0.0:8444 ssl; with the new port number to be used

Step 4. Save and restart TNMS nginx service.

7.1.2. Configure NGINX in TNMS Client

Step 1. Close TNMS Client.

Step 2. Open the file:

```
<Product_Installation_Folder>/client/application.properties
```

Step 3. Add the new port number, edit the file:

```
gctManager.NGINX_NET_SERVER_PORT=8444  
gctManager.NGINX_TNMS_SERVER_PORT=8444
```

 **NOTICE:** Ensure the line `gctManager.NGINX_TNMS_SERVER_PORT=8444` is uncommented.

Step 4. Save and open TNMS Client.

 **Note:** In the scenario of a standalone server with external mediations, the NGINX port must be updated in the server machine, using the same procedure for mediation.

Always refer to the *TNMS Communication Matrix* for the required ports to operate TNMS.

7.2. Enabling Real-time Planning and Provisioning

To enable Real-time Planning and Provisioning in TNMS or to access Real-time Planning and Provisioning during the 90 day trial license, refer to the *TNMS Administration Manual* for more information on how to manage licenses and configure the required certificates.

7.3. Enabling a Web Map Service

To enable a web map to be displayed in Real-time Planning and Provisioning, a Map Server must first be installed and then firewall ports configured to enable communicate with each TNMS Client.

After Map Server installation in TNMS go to *Administration > System Preferences > Network Map > Web Configurations >* to configure the web map service connection and preferences. See *System Preferences Online Help* for more information.

7.4. Configuring NTI Notification Service Address

If TMF CORBA NBI is installed and TNMS is running in a machine with two or more network interfaces, the Notification Service configuration must be updated by specifying the IP to which the Notification Service should bind to.

 **NOTICE:** The specified IP must be the same IP that was selected in chapter [TNMS Server and \(p. 43\) Mediation Installation \(p. 43\)](#), Step 10 .

To do so, follow the instructions below:

Step 1. Edit the file:

```
<Product_Installation_Folder>/server/NOSE/etc/jacorb.properties
```

Step 2. Update the `OAIAddr` property:

```
# IP address on multi-homed host (this gets encoded in  
# object references). Note that addresses like 127.0.0.X  
# will only be accessible from the same machine!  
# OAIAddr=1.2.3.4
```

You must uncomment the `OAIAddr` property and define it as such:

```
OAIAddr=<IP for the Notification Service>
```

Step 3. Delete the cache files:
`<Product_Data_Folder>/var/server/NOSE`

Step 4. Restart the Notification Service.
Issue:

```
# scs-service-restart nose
```

7.5. SFTP and FTP Configuration

7.5.1. Setting Up the FTP Service

The FTP service default user name is *tnms_sftp* and its home folder is the same as SFTP's: `<Product_Data_Folder>/nedata`.

Step 1. To enable FTP service on system startup, issue:

```
# systemctl enable vsftpd
```

Step 2. To start FTP service, issue:

```
# systemctl start vsftpd
```



Note: If restriction of the TNMS SFTP User (as described in [Restricting the TNMS SFTP User \(p. 61\)](#)) is required, then a dedicated FTP User is also required.

7.5.2. Restricting the TNMS SFTP User

TNMS requires the usage of an application user present in the system for information transfer between the NE network and TNMS applications. By default, this OS user is created with the ability to login and access all TNMS related folders. To enhance security, the TNMS SFTP user can be restricted to access only the application data folders and, additionally, forbidden to log into the system.

To restrict the TNMS SFTP User access to required folders only:



Note: The following instructions use:

- The default TNMS SFTP User name *tnms_sftp*
- `/nokia` to indicate the parent data directory of the `<Product_Data_Folder>`.

Step 1. In the TNMS Server machine, open the Administration Console and disable the interactive login of user *tnms_sftp*.
Issue:

```
# usermod -s /sbin/nologin tnms_sftp
```

Step 2. Change permissions on `/nokia`.
Issue:

```
# chmod 755 /nokia
```

Step 3. Edit the file `/etc/ssh/sshd_config`:
a. Comment the entry:

```
# Subsystem sftp /usr/libexec/openssh/sftp-server
```

- b. Add new entry

```
Subsystem sftp internal-sftp
```

- c. At the end of the file, add the following section:

```
Match User tnms_sftp
  ChrootDirectory /nokia
  ForceCommand internal-sftp
  X11Forwarding no
  AllowTcpForwarding no
```

- Step 4.** Apply the new settings in `sshd_config` by restarting the SSH service.

Issue:

```
# systemctl restart sshd
```

- Step 5.** Change the SFTP directory for the `tnms_sftp` user.

- a. In TNMS > System Preferences:

- i. **SFTP** tab, remove any directory before `/tnms` from the **Path** field, leaving `/tnms/nedata` only.

- ii. **External Communications** tab > **SFTP Settings**, remove any directory before `/tnms` from the **Path** field, leaving `/tnms/nedata` only.

- b. In TNMS > **NE Properties** > **SFTP Settings** tab, remove any directory before `/tnms` from the **Upload path** field, leaving `/tnms/nedata` only.

7.5.3. Setting Up the Password for TNMS, SFTP and FTP User

TNMS User

This TNMS OS user default name is `tnms`. By default the user is locked and no login is possible using this user.

If remote shell access is required for the TNMS User, the password must be set as follows:

Issue:

```
# /usr/bin/passwd tnms
```

SFTP and FTP User

The SFTP/FTP user default name is `tnms_sftp`. You must now set up its password in Linux, which for security reasons changes. When it does, the TNMS administrator must change it in the TNMS Client, according to the UMN, *Administration, SFTP* chapter.

To set up the password for the default user `tnms_sftp`, issue:

```
# /usr/bin/passwd tnms_sftp
```



NOTICE: If you installed Node Manager in a Mediation machine (only for [TNMS Special Deployment \(p. 50\)](#)), these settings must match those defined during the FTP/SFTP Configuration Step of the [TNMS Mediation Installation \(p. 51\)](#).

7.6. Configuring TNMS Service Management

The Linux service that manages TNMS must be administered as user `root`.

In alternative, TNMS user (i.e. `tnms`) can also use the `sudo` facility to manage Linux.

To configure the sudoers list and allow the TNMS user to manage the system service, proceed as follows:

Step 1. Run:

```
# cp -p $SYSTEM_INSTALL_DIR/resources/system/tnms_sudo /etc/sudoers.d
```

Step 2. Run:

```
# chown root:root /etc/sudoers.d/tnms_sudo
```

Step 3. Confirm the TNMS user is able to manage the service:

```
# sudo -l -U tnms
```



NOTICE: If the OS user configured for TNMS is not the default *tnms*, that user must be configured inside the *tnms_sudo* file, by editing the default.

8. Migrating From Other Systems

If you are migrating from another system the following features can help you setup TNMS.

8.1. Automatic Adoption of Paths

TNMS can automatically adopt new paths, changes in paths and path deletions. This option is meant to help new TNMS customers in the two following scenarios:

- **Migration** from another Network Management System (NMS).
Checking this option allows an automatic bulk adoption of all the paths in the network.
- **Parallel operation** with other systems.

Before fully migrating, you can choose to have TNMS work in parallel with the previous system. To do so, by checking this option, TNMS is able to adopt any change made by the other system, either it is the creation, change or deletion of a path.

To activate this feature go to **Main > Administration > System Preferences** and, in the **Optical Manager > Discovery** section, check **Auto-adopt NW changes done by external NMS**.

A delay must also be defined to avoid the adoption of unnecessary intermediate and inconsistent states, which would increase overall system load without any benefit. The delay is also required to maintain managed paths consistent in TNMS.

9. Update TNMS



Note: TNMS supports Rocky Linux, Oracle Linux and Red Hat Enterprise Linux (RHEL) as listed in the *Licensed Software* chapter of the *Release Notes Appendix*. All Linux instructions, hardening and commands in this manual are based on RHEL and must be adapted for Rocky Linux or Oracle Linux if required.

9.1. Upgrade from a Previous TNMS Release

To transfer your data from previous TNMS Server versions to the current TNMS Server running on Linux refer to *TNMS Upgrade Manual*.

9.2. Update the Current TNMS Release

Once the installation of TNMS 9.1 is complete, if specific components were not installed, they can be added via Update.



NOTICE: If components need to be added via Update after a Priority Update installation, copy all post installation PDTs to `<Product_Installer_Folder>/TNMS_Installer/PUs` before proceeding.

To add components to the already installed TNMS 9.1:



NOTICE: If Transcend eDNA is being added, verify that there is no version of Transcend eDNA already installed before proceeding.

Step 1. Log in as *root*.

Step 2. Go to `<Product_Installer_Folder>`, created during [TNMS Installer Integrity Validation \(p. 42\)](#) and manually assign *Execution* permissions.
Issue:

```
# cd <path>/TNMS_Installer # chmod 744 ./TNMS.bin
```

Step 3. Start the installation wizard.
Run:

```
# ./TNMS.bin
```

The installation wizard opens in **Introduction** and the complete list of installation steps is displayed on the left pane.

Step 4. A supported TNMS installation is detected and displayed in a pop up window. The detected version should match the Installer's version.

Step 5. Only displayed if Transcend Controller was not installed in the previous build. In **Server Preparation: Transcend Controller:**
Click **Enable Connection to Transcend Controller (TC)** to configure the following parameters:

- TC IP: Transcend Controller IP address.
- TC OIF Port: Transcend Controller OIF Port (Default is: 12443).
- TC Gateway Port: Transcend Controller Gateway Port (Default is 12351)



NOTICE: This is needed for the correct interworking of TNMS and Transcend Controller functionalities (for example Hot Standby, Real-time Planning and Provisioning, etc). Before proceeding, ensure that the Transcend Controller Server IP and FQDN have been configured, refer to chapter 4.1 System Hosts.

Step 6. In Server Preparation: Advisory Message:

- Enable or disable an advisory message before login.
- If enabled, personalize the advisory message in the box.

Step 7. In **Components: TNMS**, select the new components to be added. Components already installed will be unselectable.

Step 8. *Only displayed if eDNA is installed already or if selected in **Step 7**.*

In **Components: Transcend eDNA FE Servers** Insert the Transcend eDNA Frontend Server IPs.

Step 9. In **Components: Northbound Interfaces**, select the new NBIs to be added. NBIs already installed will be unselectable.

Step 10. In **Components: Network Elements**, select the new NEs to be added. NEs already installed will be unselectable.

Step 11. In **Backup Configuration**, the option to backup the database will be presented.

Step 12. A summary of the installation settings is displayed in the **Pre-Installation Summary** step. If PDTs were stored in the `<Product_Installer_Folder>\TNMS_Installer\PUs` folder, the PDT numbers will be displayed for verification. Confirm all configurations are correct, and click **Install**.

Step 13. The results of the Update are presented in the **Install Completes** step. Click Done to finish.

10. Uninstallation

Below you can find instructions on how to uninstall TNMS and its prerequisites.

10.1. Uninstalling TNMS

Before uninstalling TNMS and in case you have a standby server assigned, you must first unassign it by doing as follows **in the active server**:

- Step 1.** Select **Administration > System > Standby Server Configuration** and fill in the available fields. The address of the current standby server is filled in automatically.
- Step 2.** Verify your input and click Unassign to start the procedure. The progress and result can be followed in the configuration steps, along with the elapsed time.
- Step 3.** When the unassignment finishes, a notification pops up in the lower right corner with the status of the operation, either success or error. Alternatively, it is possible to check in System Event Log that the procedure has ended successfully.

If any error occurs, the logs can be checked in `<Product_Data_Folder>/trace/server/standby/[timestamp]/result.log`

In the **standby server**, perform the following steps:

- 1. Go to the `$SERVER_DIR/bin/standby/`, run the standby server executable file `standby-server.sh`.
- 2. Navigate to `$STANDBY_DIR`, via the Administration Console (refer to the *Administration Console* chapter in the *TNMS Administration Manual*).
- 3. Run the standby server executable file:

```
# standby-server-sh
```
- 4. In the interactive menu select **4. Start as Standalone**

To uninstall TNMS **Server** proceed as follows:

- 1. Open Administration Console.
- 2. Issue:

```
# $SYSTEM_DIR/Uninstall.sh
```
- 3. Press **Y** to confirm uninstallation.
- 4. Click **Done** on the **Uninstall Complete** panel.
- 5. To reload the profile, disconnect and reconnect the session.

To uninstall TNMS **Client**:

Refer to chapter 10.1 Uninstalling TNMS in the *TNMS Installation Manual Windows*.

 **NOTICE:** Using this process will uninstall both the TNMS Client and TNMS Server.

10.2. Uninstalling Prerequisites

Uninstalling the Database

To uninstall the database software do as follows:

- Step 1.** Login as `root`.
- Step 2.** Issue:

```
# su - oracle
```

Step 3. Issue:

```
# cd $ORACLE_HOME/deinstall  
# ./deinstall
```

The uninstaller begins.

The uninstaller requests some confirmations, such as the database details. Accept the default by pressing **Enter** or change according to your environment.

Step 4. When prompted for the Listener Name, enter **LISNER** and press **Enter**.

Step 5. When prompted for the **Database SID** enter:

- **Small/Small Plus/Medium/Large** (without Node Manager) configuration - the name of the TNMS database (TNMS is the default name).
- **Large** (with Node Manager) configuration - the name of the TNMS database (TNMS is the default name) and the name of the Node Manager database (**NMDB** is the default and mandatory name), separated by a comma, for example TNMS,**NMDB**.

Press **Enter**.

Step 6. When prompted for TNMS database modification, enter "**n**" and press **Enter**. (The details of database(s) TNMS have been discovered automatically. Do you still want to modify the details of TNMS database(s)? [n]: n).

Step 7. When prompted for continuation, enter "**y**" and press **Enter**. (Do you want to continue (y - yes, n - no)? [n]: y).

Step 8. Under the "**Clean Operation Summary**", after this message is displayed: `Oracle Universal Installer cleanup was successful`, follow the instructions in order to delete the remaining installation files.

Step 9. Additionally, as root, delete the content of the ORADATA folders (/ora1, /ora2 and /ora3).

Abbreviations

These abbreviations are intended for the entire TNMS product range and may not apply to this document in particular.

ACS

Actual Creation State

ALS

Automatic Laser Shutdown

ASON

Automatic Switched Optical Network

ASAP

Alarm Severity Assignment Profile

BFD

Bidirectional Forward Direction

BSHR

Bidirectional Self Healing Ring

CAM

Common Array Manager

CBS

Committed Burst Size

CC

Cross Connection

CCM

Continuity Check Message

CDM

Cross-domain Manager

CIR

Committed Information Rate

CFM

Connectivity Fault Management

CLI

Console Interactive

CLFI

Common Language Facility Identification

CORBA

Common Object Request Broker Architecture

CPU

Central Processing Unit

CSPF

Constrained Shortest Path First

CST

Central Standard Time

CSV

Comma-Separated Values

DB

Database

DCN

Data Communications Network

DHCP

Dynamic Host Configuration Protocol

DNS

Domain Naming Service

DP-Ring

Dedicated Protection ring

DSR

Dynamic Source Routing

DWDM

Dense Wavelength Division Multiplexing

EDU

Ethernet Demarcation Unit

ELP

Ethernet Linear Protection

EON

Embedded Optical Network

EP

Enhancement Package

FA-LSP

Forwarding Adjacency

FDR

Flight Data Recorder

FEC

Forward Error Correction

FTP

File Transfer Protocol

FRU

Field Replacement Unit

GBE

Gigabit Ethernet

GCT

GUI Cut-Through

GFP

Generic Framing Procedure Group

GM

Generic Mediator

GmbH

Gesellschaft mit beschränkter Haftung (Company with limited liability)

GMPLS

Generalized Multi-Protocol Label Switching

GMT

Greenwich Mean Time

GNE

Gateway Network Element

GPS

Global Positioning System

GUI

Graphical User Interface

HW

Hardware

IMA

Independent Management Architecture

IMN	Installation Manual
IP	Internet Protocol
JRE	Java Runtime Environment
LACP	Link Aggregation Control Protocol
LAG	Link Aggregation
LAN	Local Area Network
LAPS	Linear Automatic Protection Switching
LCT	Local Craft Terminal
LDAP	Lightweight Directory Access Protocol
LO	Low Order
LoQ	List of Quantities
LoM	List of Materials
LSP	Label Switched Path
LSR	Label Switch Router
MD	Maintenance Domain
MDI	Multiple Document Interface
MIB	Management Information Base
MLO	Multi-Layer Optimization
MLS-TP	Multiprotocol Label Switching Transport Profile
MSDE	Microsoft SQL Server Desktop Engine
MSP	Multiplex Section Protection
MTOSI	Multi Technology Operations System Interface
MVM	Multi-Vendor Mediator
NBI	Northbound Interface
NE	Network Element

NEC	NE Controller
NG	Next Generation
NIC	Network Interface Card
NMS	Network Management System
NNI	Network to Network Interface
NTFS	(Microsoft's) New Technology File System
NTI	Northbound TMF Interface
NTP	Network Time Protocol
NW	Network
OAM	Operation, Administration and Maintenance
OC	Optical Channel
ODU	Optical Data Unit - transport technology
OM	Optical Manager or Optical Management
OMS	Optical Multiplex Section
OMT	Object Model Template
OS	Operating System
OSNC	IQNOS Optical SubNetwork Connection
OPU	Optical Payload Unit - transport technology
OTDR	Optical Time Domain Reflectometer
OTN	Optical Transport Network
OTS	Optical Transport Section - transport technology
OTU	Optical Transport Unit - transport technology
PBS	Peak Burst Size
PC	Personal Computer
PCE	Path Computation Engine

PCEP

Path Computation Engine Protocol

PDF

Portable Document Format

PIR

Peak Information Rate

PMP

Performance Measurement Point

PT

Physical Trail

PTC

Planning Tool Connector

PTP

Physical Termination Point

PU

Priority Update

RAID

Redundant Array of Independent Disks

RE

Route Element

RNE

Remote Network Element

RPP

Real-time Planning and Provisioning

SBI

Southbound interface

SCSI

Small Computer System Interface

SDH

Synchronous Digital Hierarchy

SFP

Small Form-factor Pluggable

SFTP

Secure File Transfer Protocol, or Secure Shell File Transfer Protocol

SLA

Service-Level Agreement

SNC

Sub Network Connection

SNCP

SubNetwork Connection Protection

SNMP

Simple Network Management Protocol

SONET

Synchronous Optical Networking

SOR

Standard OTDR Record

SPC

Soft Permanent Connection

SQL

Structured Query Language

SRLG	Shared Risk Link Group
SSH	Secure Shell
STP	Spanning Tree Protocol
SVID	Service Virtual Local Area Network Identifier
SW	Software
TC	Topological Container or TransConnect
TCP/IP	Transport Control Protocol/Internet Protocol
TDBAK	Transient Data Backup
TL1	Transaction Language 1
TE-Link	Traffic Engineering-Link
TMN	Telecommunications Management Network
TN	TransNet
TNMS	Transcend Network Management System
TP	Terminal Point
USB	Universal Serial Bus
UMN	User Manual
UNI	User-to-Network Interface
UNI-S	User-to-Network Interface-Service
UNO	Universal Network Object
UPS	Uninterruptible Power Supply
UPSR	Unidirectional path-switched ring
VC	Virtual Container
VLAN	Virtual LAN
WAN	Wide Area Network
WDM	Wavelength Division Multiplexing

WLAN

Wireless LAN

WU

Workaround Update

XC

Cross Connection

XML

eXtended Markup Language

Glossary

These glossary entries are intended for the entire Transcend product range and may not apply to this document in particular.

@CT

@CT is a web-based craft terminal (that is, element manager) software which provides Node Manager web access to hiT 7300 network elements (NEs) in the customer network without the use of a management system. It communicates via *SNMP* with the NEs and uses the FTPS for upload/download of software or other data configuration (for example, log files).

3DES

Triple DES is the common name for the Triple Data Encryption Algorithm (TDEA or Triple DEA) symmetric-key block cipher, which applies the Data Encryption Standard (DES) cipher algorithm three times to each data block.

Actual Creation State (ACS)

Is the current state of the path which results from the accumulation of the actual creation states of the path's route elements.

Administration Console

A command line interface to issue administrative commands. In Linux root or TNMS user shell, in Windows, TNMS provided admin-console.

Advanced Encryption Standard (AES)

Is a specification for the encryption of electronic data. AES is based on a design principle known as a substitution-permutation network, and is fast in both software and hardware.

Alarm

An alarm is a management mechanism intended to inform the user that there is a standing fault condition in the system.

Alarm log

An alarm log provides a list of the alarms associated with a managed object, and provides the following information about each of the alarms:

- the identification of the affected object
- the location and the affected traffic
- the identification of the failed NE or the NE in which the failed unit resides
- the alarm severity
- the time the event occurred
- the indication of whether the alarmed event is service affecting or not

Alarm severity

Each failure is assigned a severity. The following values are used:

- indeterminate
- critical
- major
- minor
- warning
- cleared alarms
- not Existent
- not Alarmed

Alien wavelength

A wavelength that does not originate from a managed transponder or muxponder card, but is still allowed to be multiplexed into the aggregate line signal for transport as an optical channel by the system.

Automatic Laser Shutdown (ALS)

Is a technique used to automatically shut down the output power of the transmitter in case of fiber break. This is a safety feature that prevents dangerous levels of laser light from leaking out of a broken fiber, provided ALS is provisioned on both ends of the fiber pair.

Alarm Severity Assignment Profile (ASAP)

The Alarm Severity Assignment Profile is a feature that allows the management of Alarm Severity profiles in TNMS and also at the NE side.

Automatically-Switched Optical Networks (ASON)

ASON domains are built on OCh layer of hiT 7300 which have a Control Plane. The Control Plane uses network-generated signaling and routing protocols to set up or release a connection, and can restore one when it fails. ASON domains can be built up as part of the transport network. They provide the benefit of easy end-to-end provisioning, and fault and protection management. Soft permanent connections (SPCs) connect both endpoints (NE1 and NE2) within an ASON domain. If a path fails, an alternative path is automatically used.

ASON Call

A Call is a Soft Permanent Connection between two end-points (inside the same domain or between different domains) and defines the type and attributes of the connection. The establishment of a Call leads to having a path (and/or multiple alternative paths) connecting the end-points that respect the constraints and attributes defined in the Call.

Bidirectional Self- healing Ring (BSHR)

Is a telecommunications term for loop network topology, a common configuration in tele- communications transmission systems, this loop or ring is used to provide redundancy. The system consists of a ring of bidirectional links between a set of stations. In normal use, traffic is dispatched in the direction of the shortest path towards its destination. In the event of the loss of a link, or of an entire station, the two nearest surviving stations "loop back" their ends of the ring. In this way, traffic can still travel to all surviving parts of the ring, even if it has to travel "the long way round".

Capacity Planning

Capacity planning is the process of determining the capacity needed by a system to meet future needs.

Card

A card is a plug-in unit that occupies one (or multiple) shelf slots. Cards perform specific electrical and/or optical functions within an NE. Each card has a faceplate with information LEDs and, in most cases, several ports for interconnection of optical fibers and/or optical interfaces.

Card slot

A card slot is the insertion facility for a card in a shelf. Each card slot is designed for one or several particular card types. Mechanical coding elements make sure that each card can be fully inserted only into a card slot that is suitable for the given card type. Therefore, fundamental shelf equipping errors (which might cause hardware damage or fatal malfunctions) are impossible.

Ethernet Connectivity Fault Management (CFM)

Is an end-to-end per service Ethernet layer OA&M protocol. IEEE 802.1ag CFM is a service-level OA&M protocol that provides tools for detecting and isolating connectivity failures in the network. This includes proactive connectivity monitoring, fault verification and fault isolation for large Ethernet Metropolitan Area Networks (MANs) and WANs.

Committed Information Rate (CIR)

Is the guaranteed average rate (in Mbit/s) at which the information units are transferred through the port over a measurement interval.

CLFI

CLFI Codes provide a standard, mnemonic naming scheme to uniquely identify cable and transmission facilities between two standardized locations within a network. It comprises facility designation, facility type, channel/pair/time slot, location of facility terminal A and location of facility terminal Z.

Commissioning

Commissioning an network element (NE) is the process of taking an installed NE and bringing it in to an operational state. The NE commissioning phase is performed after the NE is installed and powered-up.

Controller card

NE controller cards provide the central monitoring and controlling functions of the system, as well as the MCF to operate the Q and QF Ethernet interfaces. The controller card performs the following main functions: Fault Management, Performance Management, Configuration Management, Security Management, Equipment Management, Communication Management, Software Management (performing all software downloads, uploads, and software integrity functions) and controlling the NE alarm LEDs.

Data Communication Network (DCN)

Data Communications Network is a management network for telecommunication transport systems. A DCN domain interconnects several NEs for the purpose of network management. The communication is established via the Optical Supervisory Channel (OSC) of the optical links and an Ethernet/L2 switching network implemented by the NEs.

Dense Wavelength Division Multiplexing (DWDM)

In fiber-optic communications, wavelength-division multiplexing (WDM) is a technology which multiplexes a number of optical carrier signals onto a single optical fiber by using different wavelengths (colors) of laser light, that is, simultaneously places a large number of optical signals (in C or L band) on a single optical fiber. This technique enables bidirectional communications over one strand of fiber, as well as multiplication of capacity.

Data Encryption Standard (DES)

Is a widely-used method of data encryption using a private key. DES applies a 56-bit key to each 64-bit block of data. The process can run in several modes and involves 16 rounds or operations.

Dynamic Host Configuration Protocol (DHCP)

Is a standardized networking protocol used on IP networks that dynamically configures IP addresses and other information that is needed for Internet communication. DHCP allows computers and other devices to receive an IP address automatically from a central DHCP server, reducing the need for a network administrator or a user from having to configure these settings manually.

Domain

TNMS allows you to restrict user groups to operate only a set of NEs or DCN subnets instead of the entire network. This partitioning is called a "Domain" and limits the operation on nodes outside of their partitions by assigning user groups to domains. Further, you can also assign policies to domains for further control and security, limiting the user groups to specific menu entries and actions. This arrangement is required, for example, in network centers that are responsible for maintaining only a subset of the nodes. The main purpose is security: it avoids that a login to the system grants access to the entire network. TNMS now supports the creation, modification or deletion of multiple domains, granting or restricting their accesses. By default, all NEs belong to the GLOBAL domain which cannot be modified or deleted.

Ethernet Linear Protection (ELP)

Is a protection scheme defined in the ITU-T G.8031 standard designed to protect point-to-point Ethernet paths such as VLAN based Ethernet networks. To achieve protection ELP uses two disjointed paths, a working path and a protection path, traffic is carried firstly on the active path (working path) and in case of failure, traffic is switched to the protection path. Both paths can be monitored using OAM protocols like CFM. ELP provides 1:1 bi-directional protection switching with revertive mode capabilities. ELP must first be configured at the NE side via the LCT, only then they are visible in TNMS so that you can use it in the E-LAN and E-Line service creation via the New Ethernet Service wizard. ELP is supported in specific network elements and cards only. Refer to the NE dedicated documentation for more information.

Ethernet

Ethernet is a family of frame-based computer networking technologies for LANs. It defines a number of wiring and signaling standards for the physical layer, through means of network access at the MAC/Data Link Layer, and a common addressing format.

Fault management

Fault management reports all hardware and software malfunctions within an NE, and monitors the integrity of all incoming and outgoing digital signals.

Flight Data Recorder

Flight Data Recorder (FDR) retrieves FDR files from NEs in network and provide a snapshot of the status of the Field Replacement Unit (FRU) at a given point in time.

Forward Error Correction

Forward Error Correction (FEC) or channel coding is a technique used for controlling errors in data transmission over unreliable or noisy communication channels.

File Transfer Protocol (FTP)

FTP is a network protocol used to transfer files from one computer to an NE and vice versa through the network.

Frequency

Frequency is a physical attribute of a wave (for example, an optical wave), defined as the number of wave cycles per time unit. The frequency is directly related to the wavelength.

Generalized Multi- Protocol Label Switching (GMPLS)

Is a protocol suite extending MPLS to manage further classes of interfaces and switching technologies other than packet interfaces and switching, such as time division multiplex, layer-2 switch, wavelength switch and fiber-switch.

Internet Protocol (IP)

Is the principal communications protocol in the Internet protocol suite for relaying datagrams across network boundaries. Its routing function enables internetworking, and essentially establishes the Internet.

Internet Protocol version 4 (IPv4)

Is a connectionless protocol for use on packet-switched networks. It operates on a best effort delivery model, in that it does not guarantee delivery, nor does it assure proper sequencing or avoidance of duplicate delivery. These aspects, including data integrity, are addressed by an upper layer transport protocol, such as the Transmission Control Protocol (TCP).

Job

A schedule load that must be processed by the system.

Link Aggregation Control Protocol (LACP)

Within the IEEE specification the Link Aggregation Control Protocol (LACP) provides a method to control the bundling of several physical ports together to form a single logical channel. LACP allows a network device to negotiate an automatic bundling of links by sending LACP packets to the peer (directly connected device that also implements LACP).

Link Aggregation (LAG)

Allows a bridge to treat multiple physical links between two end-points as a single logical link, referred to also as a port-channel. The feature can be used to directly connect two switches when the traffic between them requires high bandwidth and/or reliability, or to provide a higher bandwidth connection to a public network. For this purpose, all the physical links in a given port-channel must operate in full-duplex mode and at the same speed. If a physical port or the related link of a LAG fails, the traffic previously carried over the failed link automatically is switched to the remaining link(s) of the LAG (rapid reconfiguration). Bandwidth degradation is an obvious impact if the sum of throughput of the two/multiple aggregated links are higher than the throughput of the remaining link(s). Be aware that certain link failures are not always visible to both ends of a link. Link Aggregation Control Protocol (LACP) and Automatic Laser Shutdown (ALS) enabled, guarantees that both ends of a link properly detect all failures and perform the correct response. LAG groups must first be created at the NE side via the LCT, only then, they are visible in TNMS so that you can use it in the E-LAN and E-Line service creation via the New Ethernet Service wizard. LAG is supported in specific network elements and cards only. Refer to the NE dedicated documentation for more information.

Laser

A laser is a device that generates an intense narrow beam of light by stimulating the emission of photons from excited atoms or molecules.

Laser safety

Laser safety rules are a group of mechanisms and actions necessary to protect all users from harmful laser light emissions.

Local Craft network (LCT)

LCT is a client-based craft terminal (that is, element manager) software which provides access to network elements (NEs) in the customer network without the use of a management system.

Lightweight Directory Access Protocol (LDAP)

Is an application protocol for accessing and maintaining distributed directory information services over an Internet Protocol network.

Line interface

A line interface is a transponder interface that faces the line side of the link. Contrast with "client interface" which faces the client equipment side of the link. Long Haul (LH) hiT 7300 LH segment is a DWDM application characterized by a reach of more than 500 km and up to 1200 km.

Label Switched Path (LSP)

Is a path through an MPLS network, set up by a signaling protocol such as LDP, RSVPTE, BGP or CR-LDP. The path is set up based on criteria in the forwarding equivalence class (FEC).

Label switch router (LSR)

Sometimes called transit router, is a type of a router located in the middle of a Multiprotocol Label Switching (MPLS) network. It is responsible for switching the labels used to route packets. When an LSR receives a packet, it uses the label included in the packet header as an index to determine the next hop on the Label Switched Path (LSP) and a corresponding label for the packet from a look-up table. The old label is then removed from the header and replaced with the new label before the packet is routed forward.

MD5

Message-digest algorithm is a widely used cryptographic hash function producing a 128-bit (16-byte) hash value, typically expressed as a 32 digit hexadecimal number

Maintenance Association End Points (MEP)

Are points at the edge of the domain that define the boundaries and sends and receives CFM frames through the wire side (physical port) or relay function side.

Maintenance Domain (MD)

A Maintenance Domain (or OAM Domain) is a group of maintenance external and internal points (MEPs and MIPs) created directly in the NE (standard IEEE 802.1ag) that gives performance and fault values.

Management Information Base (MIB)

Is used for backup purposes where you can plan automatic upload jobs.

Multiprotocol Label Switching

Multiprotocol Label Switching (MPLS) is a mechanism in high-performance telecommunications networks that directs data from one network node to the next based on short path labels rather than long network addresses, avoiding complex lookups in a routing table. The labels identify virtual links (paths) between distant nodes rather than endpoints.

NetConf

Network Configuration Protocol (NETCONF), is an IETF network management protocol. NETCONF provides mechanisms to install, manipulate, and delete the configuration of network devices. Its operations are realized on top of a simple Remote Procedure Call (RPC) layer. The NETCONF protocol uses an Extensible Markup Language (XML) based data encoding for the configuration data as well as the protocol messages. This in turn is realized on top of the transport protocol.

Network Element (NE)

A network element (NE) is a self-contained logical unit within the network. The NE can be uniquely addressed and individually managed via software. Each NE consists of hardware and software components to perform given electrical and optical functions within the network.

Network Management

The network management layer includes all the required functions to manage the optical network in an effective and user-friendly way, such as the visualization of the network topology, creation of services, and correlation of alarms to network resources.

Network topologies

A topology of a network is defined by the list of NEs included in the network and the list of links that connect those NEs (for example, point-to-point, chain, ring, and so on).

NNI

Is an interface which specifies signaling and management functions between two networks. NNI circuit can be used for interconnection of IP (e.g. MPLS) networks.

Optical Channel

A predefined wavelength that can be used to transmit a bit stream by means of a modulated light signal.

Optical Network Node (ONN)

An ONN is an NE where the incoming channels are either dropped or routed to a line in a different direction, outgoing channels can also be added locally. Apart from multiplexing and demultiplexing an ONN NE implements optical or 3R signal regeneration and dispersion compensation.

Optical path

The path followed by an optical channel from the first multiplexer to the last demultiplexer.

Optical Transport Hierarchy (OTH)

Optical Transport Hierarchy is a transport technology for the Optical Transport Network.

Path Computation Engine Protocol (PCEP)

Implements, sets up and manages PCEP, while also notifying OM when PCEP is available or unavailable to send/receive PCEP Route messages.

Performance management

Performance monitoring and signal quality analysis provide information for detecting and alerting, a cause that could lead to a degraded performance before a failure is declared.

Peak Information Rate (PIR)

Is a burstable rate set on routers and/or switches that allows throughput overhead. Related to Committed Information Rate which is a committed rate speed guaranteed/ capped. For example, a CIR of 10 Mbit/s PIR of 12 Mbit/s allows you access to 10 Mbit/s minimum speed with burst/spike control that allows a throttle of an additional 2 Mbit/s.

Plesiochronous Digital Hierarchy (PDH)

Is a technology used to transport large quantities of data over digital transport equipment.

Pseudo-Random Binary Sequence (PRBS)

Is a known sequence of bits that can be used as a test signal to measure transmission delay and bit error rate of a channel. In this test, one port inserts the PRBS signal in the channel (source port) and another detects if the sequence was received correctly (sinkport). This kind of test is traffic affecting since the test sequence is inserted into the OPUk until the test is stopped.

Physical Trails (PT)

Trails are represented as Physical Trails (PTs). They connect two Physical Termination Points (PTP) on a physical layer rate, but can also contain non-physical layers.

Planning Tool Connector (PTC)

Interfaces Nokia TransNet/Intelligent Optical Control DWDM network planning tool.

PMP

A Performance Measurement Point is a metric represented by a set of counters for a specific point in the network. It provides data for monitoring the performance and availability of the network.

<Product_Data_Folder>

The directory defined at TNMS/Transcend Controller Installation for all software Data storage, by default /nokia/tnms or /nokia/transcend.

<Product_Installation_Folder>

The directory defined at TNMS/Transcend Controller Installation where the software is installed, by default /opt/nokia/tnms or /opt/nokia/transcend.

<Product_Installer_Folder>

The directory where the TNMS/Transcend Controller software zip was extracted before TNMS/Transcend Controller installation.



NOTICE: The <Product_Installer_Folder> must not be in the same directory as the <Product_Installation_Folder> or the <Product_Data_Folder>.

Qualitative System Requirements

Quality System Requirements are non-functional requirements that must be met by a System such as Reliability, Availability, Performance, Scalability, Security, Maintainability, Portability, etc.

Required Creation State (RCS)

Is the desired state of the path, which is set by the user upon creation.

Optical Signal to Noise Ratio (OSNR)

OSNR is the ratio of an optical signal power to the noise power in the signal.

Ring network

A ring network is a network topology in which each NE connects to exactly two other NEs, forming a circular optical path for signals (that is, a ring).

Synchronous Digital Hierarchy (SDH)

Is a standardized protocol that transfer multiple digital bit streams over optical fiber using lasers or highly coherent light from light-emitting diodes. At low transmission rates data can also be transferred via an electrical interface. The method was developed to replace the Plesiochronous Digital Hierarchy system for transporting large amounts of telephone calls and data traffic over the same fiber without synchronization problems.

Security management

Security Management controls the individual access to particular NE functions via the network management system and/or via a craft terminal, using a hierarchical security management user ID, and password concept.

State Event Machine (SEM)

In computation, a finite-state machine is event driven if the transition from one state to another is triggered by an event or a message.

Service Provisioning via NMS

Provisioning mode in hiT 7300. The core equipment is provisioned by downloading and swapping NCFs, while services are manually provisioned via the NMS. When adding new services or expanding an existing network, the relevant line cards, cross connections and internal port connections between line cards and multiplexers/demultiplexers are provisioned via the NMS.

Secure Hash Algorithm (SHA)

Is a family of cryptographic hash functions that takes an arbitrary block of data and returns a fixed-size bit string, the cryptographic hash value, such that any (accidental or intentional) change to the data will (with very high probability) change the hash value. The data to be encoded are often called the message, and the hash value is sometimes called the message digest or simply digest.

Simple Network Management Protocol (SNMP)

SNMP is used in network management systems to monitor network-attached devices for conditions that warrant administrative control. It consists of a set of standards for network management, including an application layer protocol, a database schema, and a set of data objects.

Software management

Software management performs all software downloads, uploads, and software integrity functions.

Secure Shell (SSH)

Secure Shell (SSH) Is a cryptographic network protocol for secure data communication, remote commandline login, remote command execution, and other secure network services between two networked computers that connects, via a secure channel over an insecure network, a server and a client (running SSH server and SSH client programs, respectively).

Subsystem

A subsystem is a set of shelves and cards in multicontroller NE that is controlled by a subagent. All subagents within a multicontroller NE are controlled by the master agent. Subsystem is defined for the HW only. In software, the concept is different. A major component of a system. It is made up of two or more interacting and interdependent components. Subsystems of a system interact in order to attain their own purpose(s) and the purpose(s) of the system in which they are embedded.

Synchronous Optical Networking (SONET)

Synchronous Optical Networking and Synchronous Digital Hierarchy are standardized protocols that transfer multiple digital bit streams over optical fiber using lasers or highly coherent light from light-emitting diodes.

Throughput

Throughput measures the number of work units performed in a given time unit.

Topological Container (TC)

Defines a containment relationship between other topological container and/or NEs. This means they can contain NE symbols and other TCs. The network map is always associated with one TC, which corresponds to a network view.

Tandem Connection Monitoring (TCM)

TCMs are configurable parameters (via Element Manager) of the transponders. They provide a Performance Management of all the Optical Transport Network (that is, end-to-end connection) or specific sections only and implement an Optical channel Data Unit (ODU) termination provisioned to support up to six TCM levels.

Transmission Control Protocol (TCP)

Is one of the core protocols of the Internet protocol suite (IP), and is so common that the entire suite is often called TCP/IP. TCP provides reliable, ordered, error-checked delivery of a stream of octets between programs running on computers connected to a local area network, intranet or the public Internet. It resides at the transport layer.

TL1

Transaction Language 1 (TL1) is a widely used management protocol in telecommunications. It is a cross-vendor, cross-technology man-machine language, and is widely used to manage optical (SONET) and broadband access infrastructure in North America. TL1 is used in the input and output messages that pass between Operations Systems (OSs) and Network Elements (NEs). Operations domains such as surveillance, memory administration, and access and testing define and use TL1 messages to accomplish specific functions between the OS and the NE.

TNMS

Transcend Network Management System - is a standalone application that provides a full range of network-management functions, from the transport network's physical structure and its NEs to those required for Automatically-Switched Optical Networks (ASON), SW management (also referred to as Cross-NE), Optical Management and Ethernet Management.

TNMS Core

TNMS Core is an integrated solution designed for large, medium and small size networks. It supports NEs with DWDM, OTH, SDH, PDH, Ethernet in line, star, ring and mesh network configurations. TNMS Core can be used to manage networks in the access, edge, metro, core and backbone levels.

TransNet

Planning of a hiT 7300 network is done by the Nokia TransNet tool. Nokia TransNet is a sophisticated software simulation tool developed specifically for designing and/or upgrading optical DWDM networks with hiT 7300. It runs on PCs using Microsoft Windows operating systems.

Trail Trace Identifier (TTI)

TTI is a transponder card parameter (configurable via Element Manager) of which is used to verify correct cabling or correct Tandem Connection Monitoring (TCM) configuration. The basic principle is that specific overhead bytes are reserved for Trace Messages of the user's choosing. By specifying the Actually Sent (transmitted) and the Expected (received) trace messages, the system can automatically verify that fiber connections have been made as intended. This is accomplished by comparing the expected Trace Message to that actually received. If they differ, an alarm is raised, alerting personnel of the incorrect connections.

Transponder card

A transponder card receives an optical input signal and converts it to an optical output signal suitable for DWDM multiplexing and transmission.

Transponder loopback

Loopbacks are diagnostic tests that can be activated via Element Manager. Loopbacks return the transmitted signal back to the sending device after the signal has passed across a particular link. The returned signal can then be compared to the transmitted one. Any discrepancy between the transmitted and the returned signal helps to trace faults.

User Datagram Protocol (UDP)

Is one of the core members of the Internet protocol suite (the set of network protocols used for the Internet). With UDP, computer applications can send messages, in this case referred to as datagrams, to other hosts on an Internet Protocol (IP) network without prior communications to set up special transmission channels or data paths. UDP uses a simple transmission model with a minimum of protocol mechanism. It has no handshaking dialogues, and thus exposes any unreliability of the underlying network protocol to the user's program. As this is normally IP over unreliable media, there is no guarantee of delivery, ordering or duplicate protection. UDP provides checksums for data integrity, and port numbers for addressing different functions at the source and destination of the datagram.

Ultra Long Haul (ULH)

Ultra Long Haul (ULH) hiT 7300 ULH segment is a [DWDM \(p. 70\)](#) application characterized by long path lengths of up to 1600 km.

User-to-Network Interface (UNI)

User-to-Network Interface (UNI) Is a demarcation point between the responsibility of the service provider and the responsibility of the subscriber. This is distinct from a Network to Network Interface (NNI) that defines a similar interface between provider networks

Universal Network Object (UNO)

Universal Network Object (UNO) Universal Network Objects are software NEs that can be configured and used to represent network elements which are not supported by TNMS. UNO also supports devices with restricted functionalities, for example, without supervising interfaces. They are also used to represent network services between third parties and TNMS networks.

Virtual Local Area Networks (VLAN)

In computer networking, a single layer-2 network may be partitioned to create multiple distinct broadcast domains, which are mutually isolated so that packets can only pass between them via one or more routers; such a domain is referred to as a Virtual Local Area Network, Virtual LAN or VLAN.

Wavelength

Wavelength is a physical attribute of a wave (for example, an optical wave), defined as the distance between corresponding points of two consecutive wave cycles. The wavelength is directly related to the frequency of the wave.

Wavelength Division Multiplexing (WDM)

Wavelength Division Multiplexing is a technique of multiplexing multiple optical carrier signals through a single optical fiber channel by varying the wavelengths of laser lights.

Wait to restore time (WTR)

The time in minutes that TNMS waits until it tries to switch to the working path again, assuming the Revertive option is selected.

workload Model

Representation of the typical load to be processed by the system.

eXtensible Markup Language (XML)

Is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. The design goals of XML emphasize simplicity, generality, and usability over the Internet. It is a textual data format with strong support via Unicode for the languages of the world. Although the design of XML focuses on documents, it is widely used for the representation of arbitrary data structures, for example in web services.